

Supplementary Computational Material

Frozen-Jacobian methods with predictor and memory: order, optimal selection, and fusion

Rodrigo Castro Marín

Instituto de Matemáticas, Universidad de Valparaíso, Chile

ORCID: 0000-0003-0010-7775

This supplement provides absolute timing summaries, computational-resource counts, source provenance and additional graphical representations for the numerical study. The earlier sections retain the archived float64 measurements; the final section reports the complete eight-group multiprecision family campaign, including all 496 individual timed solves. No synthetic repetition or pilot sample is introduced.

The code and archived measurement base are available in the public GitHub repository and in the software archive, version 1.0.0, DOI 10.5281/zenodo.22950468. This expanded supplement accompanies the article. Its added analyses and the new multiprecision code and measurements must not be assumed to be present in that software archive. Mathematical arguments required by the article remain independent of this computational supplement.

1 Interpretation of timing summaries

For an ordered pair A/B , let $\tilde{t}_A, \tilde{t}_B$ denote median elapsed wall times. Parentheses contain the interquartile range (IQR), in the same units. We use

$$\Delta t = 100(\tilde{t}_A/\tilde{t}_B - 1), \quad \Delta W = 100(W_A/W_B - 1).$$

Negative values favor A . A time difference is marked “clear” only if

$$|\tilde{t}_A - \tilde{t}_B| > 2(\text{IQR}_A + \text{IQR}_B).$$

This is a descriptive separation rule, not a significance test. It is applied before rounding. The detailed pair tables use six decimals in milliseconds, making the absolute medians and dispersions directly visible. Configuration ranges are not IQRs or confidence intervals.

The historical float64 CSV summaries do not include each individual timing repetition. Accordingly, this document does not reconstruct bootstrap samples or assume independent observations from medians and IQRs. Repetition-level capture in the newer software runner applies to new executions, not retrospectively to these measurements.

2 Documented environment and timing boundaries

The preserved environment files for Hammerstein and $\hat{P}_2/P_{10}$ report Windows-11-10.0.26200-SP0; AMD64 Family 25 Model 80 Stepping 0, AuthenticAMD; Python 3.13.6; NumPy 2.3.2; and SciPy 1.17.1. Both report OpenBLAS 0.3.30 with the pthreads threading layer. Two loaded BLAS libraries, associated with NumPy and SciPy, each report a 16-thread pool. This is a pool configuration captured after the runs, not a trace of simultaneous CPU usage. The reported OpenBLAS architecture string `Haswell` names its selected computational kernel and must not be interpreted as the processor model.

The two original drivers use `time.perf_counter_ns` around the complete method call. The interval includes initial copies, counters, numerical operations, residual/error norms and construction of the returned result. Problem generation, warmups, statistical aggregation and file output lie outside the interval. Their environment records confirm `maxit=20`, two warmups and 31 repetitions. Neither source forces a BLAS thread limit.

A recovered historical family-sweep driver also uses `perf_counter_ns` around its method call and declares a default maximum of 20 cycles. The preserved material does not establish the exact historical command line or BLAS environment for that sweep. For the initialization, dense external-comparator and principal fusion runs, the archived English runner documents a reproducible implementation, but its settings cannot be treated as a recovered original execution log. Deterministic cycle/resource agreement does not recover undocumented thread settings or original timing boundaries.

Table S1: Evidence levels for the timing protocols. “Not preserved” concerns the historical run record, not the availability of the newer reproducible implementation.

Suite	Environment log	Original timer source	Historical cycle cap
Hammerstein	Preserved	Complete call; ns clock	20, logged
$\hat{P}_2/P_{10}$	Preserved	Complete call; ns clock	20, logged
Family sweep	Not preserved	Retrieved driver; ns clock	20 default; invocation not preserved
I/D initialization	Not preserved	Not recovered	20 in newer runner only
Dense $P_5/M6$, $P_7/M8$	Not preserved	Not recovered	20 in newer runner only
Principal fusion	Not preserved	Not recovered	20 in newer runner; stationary mode is one fixed macrocycle

The missing historical details are not filled with settings from the current machine or from continuous-integration jobs. No fixed fraction of interpreter overhead is inferred from an assumed floating-point throughput. A new kernel-level timing study is required to separate such contributions quantitatively.

2.1 Provenance of the division-weight controls

The value $\kappa = 1$ is the algebraic reference convention. The controls 1.2, 1.298 and 1.7 retain historical sensitivity choices; they are not universal hardware constants. The recovered internal record mentions published precedents for the latter two without identifying an exact primary source; no historical attribution is made here on that basis.

The scalar calibration record reports 3.23511 for throughput (eight independent chains) and 4.45444 for latency (one dependent chain), in `float64`, with Python 3.13.6, NumPy 2.3.2 and Numba 0.67.0 on the AMD/Windows platform stated above. The retained summary also reports 0.314174 ns per multiplication, 1.016274 ns per division and an IQR of 0.022895 for the throughput ratio. A ratio of summarized times need not equal the summarized paired ratios. Batch-level calibration files were not recovered in this documentary review, so those summary values are not independently re-estimated here. They are used only as sensitivity controls. They neither calibrate LU execution time nor provide weights for arbitrary-precision arithmetic.

3 Nominal configurations and fixed-block paths

A nominal configuration includes field, dimension, initial point, method and tolerance. Changing the tolerance can leave a fixed-block trajectory unchanged. For a fixed ordered method pair, we group configurations by field, dimension, initial distance and the two observed cycle counts. Since

the tolerance only controls the residual stop at block boundaries, equal signatures correspond to the same algorithmic path under the stated deterministic protocol. This grouping is inferred from the protocol and archived counters; it does not assert statistical independence or recover unrecorded trajectories.

Table S2: Nominal pairs versus paired fixed-block path signatures. Different initial distances are retained as different signatures.

Suite	Nominal pairs	Path signatures	Both finish in one cycle
$P_5/M6$	45	20	36
$P_7/M8$	45	16	44
Fusion, end-to-end	72	24	72
$\hat{P}_2/P_{10}$	18	6	18
Hammerstein, $P_5/M6$	3	1	0
Hammerstein, $P_7/M8$	3	2	0

A repeated trajectory can still be timed repeatedly. The limitation is on claims of problem diversity or independence, not on the existence of the recorded timing repetitions. In particular, the 72 end-to-end fusion pairs do not constitute 72 independent nonlinear problems.

4 Complete external-comparator timings

Tables S3 and S4 contain all three tolerances, rather than only the 10^{-12} subset. They are direct reorganizations of the archived CSV rows and preserve both favorable and unfavorable outcomes.

Table S3: P_5 versus $M6$. Medians and IQRs in parentheses are in milliseconds. The final column applies the descriptive separation rule to unrounded values.

Field	n	δ	TOL	Cycles	P_5 time (IQR)	$M6$ time (IQR)	Δt [%]	Clear
H_1	20	0.10	10^{-8}	1/1	0.126200 (0.006200)	0.124900 (0.012550)	+1.04	no
H_1	20	0.10	10^{-10}	1/1	0.127600 (0.010650)	0.125900 (0.011300)	+1.35	no
H_1	20	0.10	10^{-12}	1/1	0.125800 (0.006400)	0.125200 (0.007500)	+0.48	no
H_1	20	0.30	10^{-8}	1/1	0.126000 (0.004850)	0.125000 (0.008100)	+0.80	no
H_1	20	0.30	10^{-10}	2/2	0.251900 (0.010900)	0.235600 (0.018250)	+6.92	no
H_1	20	0.30	10^{-12}	2/2	0.247800 (0.010700)	0.242800 (0.017500)	+2.06	no
H_1	20	0.60	10^{-8}	2/2	0.250700 (0.010750)	0.236900 (0.017350)	+5.83	no
H_1	20	0.60	10^{-10}	2/2	0.256800 (0.013550)	0.241400 (0.010750)	+6.38	no
H_1	20	0.60	10^{-12}	2/2	0.249500 (0.015100)	0.242300 (0.012400)	+2.97	no
H_2	50	0.10	10^{-8}	1/1	0.170600 (0.012950)	0.172900 (0.011700)	-1.33	no
H_2	50	0.10	10^{-10}	1/1	0.173400 (0.013750)	0.171500 (0.013250)	+1.11	no
H_2	50	0.10	10^{-12}	1/1	0.175200 (0.010400)	0.174700 (0.012350)	+0.29	no
H_2	50	0.30	10^{-8}	1/1	0.177600 (0.019700)	0.178700 (0.022800)	-0.62	no
H_2	50	0.30	10^{-10}	1/1	0.174300 (0.011600)	0.174400 (0.013600)	-0.06	no
H_2	50	0.30	10^{-12}	1/1	0.173000 (0.013600)	0.173100 (0.014100)	-0.06	no
H_2	50	0.60	10^{-8}	1/1	0.175300 (0.010300)	0.172900 (0.005100)	+1.39	no
H_2	50	0.60	10^{-10}	1/1	0.174900 (0.015700)	0.174600 (0.021100)	+0.17	no
H_2	50	0.60	10^{-12}	2/2	0.351400 (0.026250)	0.336200 (0.018550)	+4.52	no
H_3	100	0.10	10^{-8}	1/1	0.235500 (0.018600)	0.247100 (0.014450)	-4.69	no
H_3	100	0.10	10^{-10}	1/1	0.251200 (0.022850)	0.251200 (0.018750)	+0.00	no
H_3	100	0.10	10^{-12}	1/1	0.244200 (0.038850)	0.250600 (0.031000)	-2.55	no
H_3	100	0.30	10^{-8}	1/1	0.241400 (0.009400)	0.252200 (0.007800)	-4.28	no
H_3	100	0.30	10^{-10}	1/1	0.245700 (0.027400)	0.251900 (0.018200)	-2.46	no
H_3	100	0.30	10^{-12}	1/1	0.241100 (0.012650)	0.249900 (0.020700)	-3.52	no
H_3	100	0.60	10^{-8}	1/1	0.244000 (0.027750)	0.253600 (0.026550)	-3.79	no
H_3	100	0.60	10^{-10}	1/1	0.242700 (0.018150)	0.257400 (0.017800)	-5.71	no
H_3	100	0.60	10^{-12}	2/2	0.477800 (0.013200)	0.490900 (0.027550)	-2.67	no
H_4	150	0.10	10^{-8}	1/1	0.539700 (0.041100)	0.563300 (0.061800)	-4.19	no
H_4	150	0.10	10^{-10}	1/1	0.570800 (0.065450)	0.571700 (0.075200)	-0.16	no
H_4	150	0.10	10^{-12}	1/1	0.549500 (0.030100)	0.570400 (0.069900)	-3.66	no
H_4	150	0.30	10^{-8}	1/1	0.575900 (0.102200)	0.607500 (0.091800)	-5.20	no
H_4	150	0.30	10^{-10}	1/1	0.576500 (0.064050)	0.611600 (0.111150)	-5.74	no
H_4	150	0.30	10^{-12}	1/1	0.550500 (0.035550)	0.577200 (0.055150)	-4.63	no
H_4	150	0.60	10^{-8}	1/1	0.559600 (0.052200)	0.572600 (0.056500)	-2.27	no
H_4	150	0.60	10^{-10}	1/1	0.549700 (0.044600)	0.578200 (0.091200)	-4.93	no
H_4	150	0.60	10^{-12}	2/2	1.138200 (0.117550)	1.140200 (0.106400)	-0.18	no
H_5	200	0.10	10^{-8}	1/1	1.023200 (0.067650)	1.057900 (0.066200)	-3.28	no
H_5	200	0.10	10^{-10}	1/1	1.016600 (0.078600)	1.039500 (0.162700)	-2.20	no
H_5	200	0.10	10^{-12}	1/1	1.037800 (0.084350)	1.082100 (0.127200)	-4.09	no
H_5	200	0.30	10^{-8}	1/1	1.074500 (0.130000)	1.112800 (0.080750)	-3.44	no
H_5	200	0.30	10^{-10}	1/1	1.062200 (0.128700)	1.105600 (0.138400)	-3.93	no
H_5	200	0.30	10^{-12}	1/1	1.076000 (0.074200)	1.116400 (0.080350)	-3.62	no
H_5	200	0.60	10^{-8}	1/1	1.094100 (0.081100)	1.141300 (0.152950)	-4.14	no
H_5	200	0.60	10^{-10}	1/1	1.084600 (0.096750)	1.117800 (0.107850)	-2.97	no
H_5	200	0.60	10^{-12}	2/2	2.058800 (0.130500)	2.121700 (0.212850)	-2.96	no

Table S4: P_7 versus $M8$. Medians and IQRs in parentheses are in milliseconds. The final column applies the descriptive separation rule to unrounded values.

Field	n	δ	TOL	Cycles	P_7 time (IQR)	$M8$ time (IQR)	Δt [%]	Clear
H_1	20	0.10	10^{-8}	1/1	0.157800 (0.010500)	0.123200 (0.010600)	+28.08	no
H_1	20	0.10	10^{-10}	1/1	0.153100 (0.008050)	0.122300 (0.011650)	+25.18	no
H_1	20	0.10	10^{-12}	1/1	0.153700 (0.008450)	0.120200 (0.005000)	+27.87	yes
H_1	20	0.30	10^{-8}	1/1	0.154600 (0.007100)	0.124400 (0.011400)	+24.28	no
H_1	20	0.30	10^{-10}	1/1	0.153900 (0.004000)	0.120500 (0.006150)	+27.72	yes
H_1	20	0.30	10^{-12}	1/1	0.155800 (0.010200)	0.122000 (0.010800)	+27.70	no
H_1	20	0.60	10^{-8}	1/1	0.155000 (0.007700)	0.122200 (0.012750)	+26.84	no
H_1	20	0.60	10^{-10}	1/1	0.156800 (0.010000)	0.121900 (0.005150)	+28.63	yes
H_1	20	0.60	10^{-12}	2/2	0.305700 (0.010250)	0.227400 (0.008850)	+34.43	yes
H_2	50	0.10	10^{-8}	1/1	0.210900 (0.011350)	0.180900 (0.012500)	+16.58	no
H_2	50	0.10	10^{-10}	1/1	0.207100 (0.007950)	0.177500 (0.006350)	+16.68	yes
H_2	50	0.10	10^{-12}	1/1	0.204700 (0.009650)	0.177400 (0.015500)	+15.39	no
H_2	50	0.30	10^{-8}	1/1	0.209800 (0.016000)	0.180000 (0.012300)	+16.56	no
H_2	50	0.30	10^{-10}	1/1	0.206700 (0.012250)	0.177700 (0.004100)	+16.32	no
H_2	50	0.30	10^{-12}	1/1	0.212800 (0.013900)	0.180800 (0.016900)	+17.70	no
H_2	50	0.60	10^{-8}	1/1	0.209900 (0.011550)	0.179500 (0.010350)	+16.94	no
H_2	50	0.60	10^{-10}	1/1	0.206800 (0.017600)	0.183000 (0.011400)	+13.01	no
H_2	50	0.60	10^{-12}	1/1	0.210300 (0.011550)	0.180100 (0.010550)	+16.77	no
H_3	100	0.10	10^{-8}	1/1	0.283700 (0.007300)	0.285700 (0.014050)	-0.70	no
H_3	100	0.10	10^{-10}	1/1	0.290600 (0.015850)	0.286600 (0.011350)	+1.40	no
H_3	100	0.10	10^{-12}	1/1	0.285000 (0.017600)	0.284400 (0.014850)	+0.21	no
H_3	100	0.30	10^{-8}	1/1	0.282700 (0.014750)	0.285300 (0.013100)	-0.91	no
H_3	100	0.30	10^{-10}	1/1	0.282900 (0.015600)	0.289000 (0.016600)	-2.11	no
H_3	100	0.30	10^{-12}	1/1	0.287300 (0.016250)	0.285600 (0.011550)	+0.60	no
H_3	100	0.60	10^{-8}	1/1	0.286600 (0.011550)	0.291000 (0.017050)	-1.51	no
H_3	100	0.60	10^{-10}	1/1	0.287400 (0.014150)	0.294800 (0.016750)	-2.51	no
H_3	100	0.60	10^{-12}	1/1	0.286900 (0.012350)	0.288100 (0.016750)	-0.42	no
H_4	150	0.10	10^{-8}	1/1	0.586000 (0.037850)	0.727800 (0.069200)	-19.48	no
H_4	150	0.10	10^{-10}	1/1	0.592700 (0.052250)	0.754600 (0.059700)	-21.46	no
H_4	150	0.10	10^{-12}	1/1	0.583900 (0.039150)	0.736100 (0.049500)	-20.68	no
H_4	150	0.30	10^{-8}	1/1	0.575700 (0.044050)	0.727300 (0.114050)	-20.84	no
H_4	150	0.30	10^{-10}	1/1	0.580500 (0.021450)	0.726300 (0.037500)	-20.07	yes
H_4	150	0.30	10^{-12}	1/1	0.610700 (0.074400)	0.772900 (0.066300)	-20.99	no
H_4	150	0.60	10^{-8}	1/1	0.592900 (0.028700)	0.746900 (0.100300)	-20.62	no
H_4	150	0.60	10^{-10}	1/1	0.593400 (0.032700)	0.740500 (0.044050)	-19.86	no
H_4	150	0.60	10^{-12}	1/1	0.594700 (0.047100)	0.734800 (0.104700)	-19.07	no
H_5	200	0.10	10^{-8}	1/1	1.074600 (0.050050)	1.632400 (0.113350)	-34.17	yes
H_5	200	0.10	10^{-10}	1/1	1.061900 (0.051000)	1.624100 (0.073750)	-34.62	yes
H_5	200	0.10	10^{-12}	1/1	1.156800 (0.098350)	1.798600 (0.197550)	-35.68	yes
H_5	200	0.30	10^{-8}	1/1	1.052600 (0.065050)	1.606900 (0.084250)	-34.49	yes
H_5	200	0.30	10^{-10}	1/1	1.088200 (0.069750)	1.652300 (0.100200)	-34.14	yes
H_5	200	0.30	10^{-12}	1/1	1.084200 (0.081850)	1.650400 (0.119850)	-34.31	yes
H_5	200	0.60	10^{-8}	1/1	1.102300 (0.123000)	1.677600 (0.146600)	-34.29	yes
H_5	200	0.60	10^{-10}	1/1	1.077100 (0.036250)	1.651500 (0.064550)	-34.78	yes
H_5	200	0.60	10^{-12}	1/1	1.075800 (0.035150)	1.645000 (0.135700)	-34.60	yes

5 Fusion and nearby-order comparison

Table S5 times one prepared stationary macrocycle; it is not an end-to-end tolerance test. Tables S6 and S7 include delayed initialization and the terminal residual evaluation. Throughout these measurements $\hat{P}_2 = \hat{P}_{2,4}$; no new measurements for $q = 2$ or $q = 3$ are supplied here.

Table S5: $\widehat{P}_2$ versus $P_2 \circ P_2$. Medians and IQRs in parentheses are in milliseconds. The final column applies the descriptive separation rule to unrounded values.

Field	n	δ	TOL	Cycles	$\widehat{P}_2$ time (IQR)	$P_2 \circ P_2$ time (IQR)	Δt [%]	Clear
G_1	40	—	—	1/1	0.242600 (0.014200)	0.154400 (0.004900)	+57.12	yes
G_1	55	—	—	1/1	0.270800 (0.012300)	0.184800 (0.012850)	+46.54	yes
G_1	100	—	—	1/1	0.381000 (0.042850)	0.298000 (0.044550)	+27.85	no
G_1	200	—	—	1/1	1.699900 (0.152500)	2.078800 (0.246800)	-18.23	no
G_5	40	—	—	1/1	0.264200 (0.005400)	0.176100 (0.006350)	+50.03	yes
G_5	55	—	—	1/1	0.294200 (0.008900)	0.206600 (0.009200)	+42.40	yes
G_5	100	—	—	1/1	0.399400 (0.017100)	0.314000 (0.012550)	+27.20	yes
G_5	200	—	—	1/1	1.654400 (0.184700)	2.048300 (0.292050)	-19.23	no

Table S6: $\widehat{P}_2$ versus $P_2 \circ P_2$. Medians and IQRs in parentheses are in milliseconds. The final column applies the descriptive separation rule to unrounded values.

Field	n	δ	TOL	Cycles	$\widehat{P}_2$ time (IQR)	$P_2 \circ P_2$ time (IQR)	Δt [%]	Clear
G_1	40	0.10	10^{-8}	1/1	0.231500 (0.030150)	0.176800 (0.016400)	+30.94	no
G_1	40	0.10	10^{-10}	1/1	0.219500 (0.046850)	0.165800 (0.014100)	+32.39	no
G_1	40	0.10	10^{-12}	1/1	0.221900 (0.039150)	0.168100 (0.014250)	+32.00	no
G_1	40	0.30	10^{-8}	1/1	0.219200 (0.016150)	0.167400 (0.011700)	+30.94	no
G_1	40	0.30	10^{-10}	1/1	0.218900 (0.015750)	0.164100 (0.009050)	+33.39	yes
G_1	40	0.30	10^{-12}	1/1	0.224400 (0.065400)	0.166000 (0.018750)	+35.18	no
G_1	40	0.60	10^{-8}	1/1	0.217600 (0.017950)	0.165000 (0.065200)	+31.88	no
G_1	40	0.60	10^{-10}	1/1	0.219300 (0.008400)	0.166900 (0.011100)	+31.40	yes
G_1	40	0.60	10^{-12}	1/1	0.218700 (0.012000)	0.166600 (0.010350)	+31.27	yes
G_1	55	0.10	10^{-8}	1/1	0.254400 (0.019500)	0.202800 (0.021150)	+25.44	no
G_1	55	0.10	10^{-10}	1/1	0.249500 (0.009550)	0.199600 (0.006950)	+25.00	yes
G_1	55	0.10	10^{-12}	1/1	0.252200 (0.013150)	0.200800 (0.007600)	+25.60	yes
G_1	55	0.30	10^{-8}	1/1	0.251200 (0.009250)	0.202100 (0.011550)	+24.29	yes
G_1	55	0.30	10^{-10}	1/1	0.257200 (0.016300)	0.203300 (0.010500)	+26.51	yes
G_1	55	0.30	10^{-12}	1/1	0.251200 (0.008700)	0.202400 (0.006950)	+24.11	yes
G_1	55	0.60	10^{-8}	1/1	0.252200 (0.016550)	0.208000 (0.011350)	+21.25	no
G_1	55	0.60	10^{-10}	1/1	0.255200 (0.016350)	0.202700 (0.013850)	+25.90	no
G_1	55	0.60	10^{-12}	1/1	0.253900 (0.012850)	0.202800 (0.015000)	+25.20	no
G_1	100	0.10	10^{-8}	1/1	0.354600 (0.027350)	0.308400 (0.032750)	+14.98	no
G_1	100	0.10	10^{-10}	1/1	0.343800 (0.024400)	0.304200 (0.014800)	+13.02	no
G_1	100	0.10	10^{-12}	1/1	0.339900 (0.016750)	0.304200 (0.014700)	+11.74	no
G_1	100	0.30	10^{-8}	1/1	0.340500 (0.020850)	0.301600 (0.011850)	+12.90	no
G_1	100	0.30	10^{-10}	1/1	0.340000 (0.012800)	0.299200 (0.008000)	+13.64	no
G_1	100	0.30	10^{-12}	1/1	0.344300 (0.038450)	0.309200 (0.021700)	+11.35	no
G_1	100	0.60	10^{-8}	1/1	0.346000 (0.015450)	0.300900 (0.009950)	+14.99	no
G_1	100	0.60	10^{-10}	1/1	0.345000 (0.026300)	0.304100 (0.020850)	+13.45	no
G_1	100	0.60	10^{-12}	1/1	0.338700 (0.014350)	0.300700 (0.010000)	+12.64	no
G_1	200	0.10	10^{-8}	1/1	1.298200 (0.079250)	1.792200 (0.119800)	-27.56	yes
G_1	200	0.10	10^{-10}	1/1	1.241600 (0.054700)	1.770500 (0.126350)	-29.87	yes
G_1	200	0.10	10^{-12}	1/1	1.269800 (0.110050)	1.836900 (0.117500)	-30.87	yes
G_1	200	0.30	10^{-8}	1/1	1.258300 (0.073900)	1.773500 (0.153300)	-29.05	yes
G_1	200	0.30	10^{-10}	1/1	1.250500 (0.095800)	1.768800 (0.132650)	-29.30	yes
G_1	200	0.30	10^{-12}	1/1	1.268300 (0.094700)	1.776700 (0.107450)	-28.61	yes
G_1	200	0.60	10^{-8}	1/1	1.259900 (0.126950)	1.789100 (0.124550)	-29.58	yes
G_1	200	0.60	10^{-10}	1/1	1.243200 (0.076050)	1.766100 (0.123000)	-29.61	yes
G_1	200	0.60	10^{-12}	1/1	1.272900 (0.074750)	1.820800 (0.181250)	-30.09	yes
G_5	40	0.10	10^{-8}	1/1	0.237800 (0.015450)	0.187000 (0.014750)	+27.17	no
G_5	40	0.10	10^{-10}	1/1	0.245900 (0.020800)	0.186100 (0.002250)	+32.13	yes
G_5	40	0.10	10^{-12}	1/1	0.239600 (0.013450)	0.188000 (0.015000)	+27.45	no
G_5	40	0.30	10^{-8}	1/1	0.238400 (0.017850)	0.187200 (0.014600)	+27.35	no

Continued on the next page

Table S6 (continued)

Field	n	δ	TOL	Cycles	$\hat{P}_2$ time (IQR)	$P_2 \circ P_2$ time (IQR)	Δt [%]	Clear
G_5	40	0.30	10^{-10}	1/1	0.240900 (0.021350)	0.195500 (0.021350)	+23.22	no
G_5	40	0.30	10^{-12}	1/1	0.239300 (0.017350)	0.185700 (0.012350)	+28.86	no
G_5	40	0.60	10^{-8}	1/1	0.243500 (0.026550)	0.187700 (0.010800)	+29.73	no
G_5	40	0.60	10^{-10}	1/1	0.240900 (0.014150)	0.187400 (0.019950)	+28.55	no
G_5	40	0.60	10^{-12}	1/1	0.241500 (0.020700)	0.187100 (0.021550)	+29.08	no
G_5	55	0.10	10^{-8}	1/1	0.282400 (0.015150)	0.224600 (0.027400)	+25.73	no
G_5	55	0.10	10^{-10}	1/1	0.282900 (0.042850)	0.236700 (0.027550)	+19.52	no
G_5	55	0.10	10^{-12}	1/1	0.275600 (0.015200)	0.222300 (0.013300)	+23.98	no
G_5	55	0.30	10^{-8}	1/1	0.272500 (0.017100)	0.230400 (0.023050)	+18.27	no
G_5	55	0.30	10^{-10}	1/1	0.274100 (0.023350)	0.223700 (0.012750)	+22.53	no
G_5	55	0.30	10^{-12}	1/1	0.276900 (0.011950)	0.223000 (0.016200)	+24.17	no
G_5	55	0.60	10^{-8}	1/1	0.273300 (0.005950)	0.225700 (0.008750)	+21.09	yes
G_5	55	0.60	10^{-10}	1/1	0.273900 (0.013500)	0.224100 (0.013400)	+22.22	no
G_5	55	0.60	10^{-12}	1/1	0.273500 (0.011550)	0.223600 (0.013350)	+22.32	yes
G_5	100	0.10	10^{-8}	1/1	0.367900 (0.021100)	0.326700 (0.012700)	+12.61	no
G_5	100	0.10	10^{-10}	1/1	0.363600 (0.025200)	0.330800 (0.052450)	+9.92	no
G_5	100	0.10	10^{-12}	1/1	0.362000 (0.013750)	0.332200 (0.014550)	+8.97	no
G_5	100	0.30	10^{-8}	1/1	0.368600 (0.017000)	0.333200 (0.032850)	+10.62	no
G_5	100	0.30	10^{-10}	1/1	0.373200 (0.032500)	0.341200 (0.027300)	+9.38	no
G_5	100	0.30	10^{-12}	1/1	0.364300 (0.021150)	0.335000 (0.022200)	+8.75	no
G_5	100	0.60	10^{-8}	1/1	0.368800 (0.025400)	0.334500 (0.017200)	+10.25	no
G_5	100	0.60	10^{-10}	1/1	0.372500 (0.022250)	0.326300 (0.020150)	+14.16	no
G_5	100	0.60	10^{-12}	1/1	0.371800 (0.021800)	0.335800 (0.016750)	+10.72	no
G_5	200	0.10	10^{-8}	1/1	1.268000 (0.062950)	1.818500 (0.099100)	-30.27	yes
G_5	200	0.10	10^{-10}	1/1	1.304000 (0.140100)	1.830800 (0.161800)	-28.77	no
G_5	200	0.10	10^{-12}	1/1	1.306800 (0.111400)	1.804000 (0.089250)	-27.56	yes
G_5	200	0.30	10^{-8}	1/1	1.297400 (0.089350)	1.802200 (0.121550)	-28.01	yes
G_5	200	0.30	10^{-10}	1/1	1.283900 (0.069600)	1.852200 (0.127300)	-30.68	yes
G_5	200	0.30	10^{-12}	1/1	1.312000 (0.112050)	1.822900 (0.087100)	-28.03	yes
G_5	200	0.60	10^{-8}	1/1	1.299600 (0.113600)	1.871700 (0.125150)	-30.57	yes
G_5	200	0.60	10^{-10}	1/1	1.295100 (0.088600)	1.838600 (0.133650)	-29.56	yes
G_5	200	0.60	10^{-12}	1/1	1.302900 (0.132350)	1.822200 (0.132600)	-28.50	no

Table S7: $\hat{P}_2$ versus P_{10} . Medians and IQRs in parentheses are in milliseconds. The final column applies the descriptive separation rule to unrounded values.

Field	n	δ	TOL	Cycles	$\hat{P}_2$ time (IQR)	P_{10} time (IQR)	Δt [%]	Clear
H_3	100	0.10	10^{-8}	1/1	0.364000 (0.034250)	0.351300 (0.034750)	+3.62	no
H_3	100	0.10	10^{-10}	1/1	0.355900 (0.021100)	0.354600 (0.020900)	+0.37	no
H_3	100	0.10	10^{-12}	1/1	0.352400 (0.022850)	0.346100 (0.019550)	+1.82	no
H_3	100	0.30	10^{-8}	1/1	0.353000 (0.028750)	0.359600 (0.025800)	-1.84	no
H_3	100	0.30	10^{-10}	1/1	0.355200 (0.021150)	0.345800 (0.025750)	+2.72	no
H_3	100	0.30	10^{-12}	1/1	0.361100 (0.027350)	0.354700 (0.019300)	+1.80	no
H_3	100	0.60	10^{-8}	1/1	0.359800 (0.025750)	0.359500 (0.021750)	+0.08	no
H_3	100	0.60	10^{-10}	1/1	0.346200 (0.033000)	0.354000 (0.021200)	-2.20	no
H_3	100	0.60	10^{-12}	1/1	0.355000 (0.027100)	0.350400 (0.018350)	+1.31	no
H_5	200	0.10	10^{-8}	1/1	1.226400 (0.055950)	1.227800 (0.096850)	-0.11	no
H_5	200	0.10	10^{-10}	1/1	1.227300 (0.120850)	1.236700 (0.103850)	-0.76	no
H_5	200	0.10	10^{-12}	1/1	1.284900 (0.139400)	1.267000 (0.151100)	+1.41	no
H_5	200	0.30	10^{-8}	1/1	1.225900 (0.078150)	1.277900 (0.113700)	-4.07	no
H_5	200	0.30	10^{-10}	1/1	1.229900 (0.086350)	1.250800 (0.099000)	-1.67	no
H_5	200	0.30	10^{-12}	1/1	1.244600 (0.088150)	1.234700 (0.061050)	+0.80	no
H_5	200	0.60	10^{-8}	1/1	1.230900 (0.115350)	1.242200 (0.104050)	-0.91	no
H_5	200	0.60	10^{-10}	1/1	1.241200 (0.088200)	1.259400 (0.109600)	-1.45	no
H_5	200	0.60	10^{-12}	1/1	1.255400 (0.142250)	1.248200 (0.110650)	+0.58	no

Work-column provenance for G_5 . The historical principal-fusion CSV work columns retain $\mu_1 = 2 + 18.2/n$ at $\kappa = 1$, whereas the canonical formula is $2 + (17 + \kappa)/n$. This difference concerns calculated work, not measured times or counters. The archived values are not silently overwritten. The separately recomputed canonical-work comparison is retained with the source evidence; the elapsed times and all separation decisions above are unaffected. Relative-work values reproduced in the article retain their displayed rounding.

6 External Hammerstein benchmark

Tables S8 and S9 use the eight-node discretization and the common initial vector specified in the article. The dash in the distance column means that this benchmark uses its specified initial point rather than the dense-field distance parameter.

Table S8: P_5 versus $M6$. Medians and IQRs in parentheses are in milliseconds. The final column applies the descriptive separation rule to unrounded values.

Field	n	δ	TOL	Cycles	P_5 time (IQR)	$M6$ time (IQR)	Δt [%]	Clear
Hammerstein	8	—	10^{-8}	2/2	0.180300 (0.008250)	0.169200 (0.008400)	+6.56	no
Hammerstein	8	—	10^{-10}	2/2	0.179500 (0.004250)	0.168700 (0.002900)	+6.40	no
Hammerstein	8	—	10^{-12}	2/2	0.179900 (0.004750)	0.169700 (0.012450)	+6.01	no

Table S9: P_7 versus $M8$. Medians and IQRs in parentheses are in milliseconds. The final column applies the descriptive separation rule to unrounded values.

Field	n	δ	TOL	Cycles	P_7 time (IQR)	$M8$ time (IQR)	Δt [%]	Clear
Hammerstein	8	—	10^{-8}	1/2	0.115100 (0.003300)	0.167200 (0.011300)	-31.16	yes
Hammerstein	8	—	10^{-10}	2/2	0.229600 (0.011350)	0.167100 (0.009250)	+37.40	yes
Hammerstein	8	—	10^{-12}	2/2	0.228500 (0.008600)	0.166400 (0.008800)	+37.32	yes

7 Family sweep and initialization

The principal family comparison selects $P_{N_P^*}$ and $S_{m_S^*}$ independently from the stationary model. Table S10 preserves the wider sweep as a sensitivity diagnostic, not a new primary comparison between arbitrary neighbors. The minimum observed timing median need not coincide with the stationary optimum. For a one-cycle delayed run, P_j and S_j execute the same block.

Table S10: Complete family sweep at $\text{TOL} = 10^{-12}$. Times are in milliseconds. These are descriptive measurements across indices; the primary family comparison uses the independently selected stationary optima.

Field	δ	Method	Cycles	Median	IQR	F	J	LU	Solves
H_1	0.10	P_2	2	0.136400	0.004600	5	2	2	5
H_1	0.10	P_3	2	0.168200	0.004900	7	2	2	7
H_1	0.10	P_4	2	0.201100	0.010800	9	2	2	9
H_1	0.10	P_5	1	0.118800	0.004700	6	1	1	5
H_1	0.10	P_6	1	0.134800	0.003150	7	1	1	6
H_1	0.10	P_7	1	0.151100	0.002000	8	1	1	7
H_1	0.10	P_8	1	0.166800	0.003950	9	1	1	8
H_1	0.10	P_9	1	0.183100	0.005700	10	1	1	9
H_1	0.10	P_{10}	1	0.200000	0.004650	11	1	1	10
H_1	0.10	P_{11}	1	0.215600	0.007250	12	1	1	11
H_1	0.10	P_{12}	1	0.232500	0.009000	13	1	1	12
H_1	0.10	P_{13}	1	0.247200	0.004000	14	1	1	13
H_1	0.10	P_{14}	1	0.262800	0.005550	15	1	1	14
H_1	0.10	P_{15}	1	0.281400	0.011250	16	1	1	15
H_1	0.10	P_{16}	1	0.295500	0.011900	17	1	1	16
H_1	0.10	P_{17}	1	0.310500	0.009650	18	1	1	17
H_1	0.10	P_{18}	1	0.327400	0.006950	19	1	1	18
H_1	0.10	P_{19}	1	0.342700	0.011250	20	1	1	19
H_1	0.10	P_{20}	1	0.359500	0.008150	21	1	1	20
H_1	0.10	S_1	3	0.136700	0.003950	4	3	3	3
H_1	0.10	S_2	2	0.129300	0.006950	5	2	2	4
H_1	0.10	S_3	2	0.161900	0.003200	7	2	2	6
H_1	0.10	S_4	2	0.194100	0.006450	9	2	2	8
H_1	0.10	S_5	1	0.120100	0.004150	6	1	1	5
H_1	0.10	S_6	1	0.136500	0.004750	7	1	1	6
H_1	0.10	S_7	1	0.151100	0.003950	8	1	1	7
H_1	0.10	S_8	1	0.168100	0.004550	9	1	1	8
H_1	0.10	S_9	1	0.183100	0.004650	10	1	1	9
H_1	0.10	S_{10}	1	0.198900	0.003350	11	1	1	10
H_1	0.10	S_{11}	1	0.215900	0.004650	12	1	1	11
H_1	0.10	S_{12}	1	0.233400	0.007150	13	1	1	12
H_1	0.10	S_{13}	1	0.248300	0.005450	14	1	1	13
H_1	0.10	S_{14}	1	0.264200	0.009650	15	1	1	14
H_1	0.10	S_{15}	1	0.281400	0.010100	16	1	1	15
H_1	0.10	S_{16}	1	0.296500	0.007350	17	1	1	16
H_1	0.10	S_{17}	1	0.312500	0.008800	18	1	1	17
H_1	0.10	S_{18}	1	0.329600	0.012850	19	1	1	18
H_1	0.10	S_{19}	1	0.345700	0.006600	20	1	1	19
H_1	0.10	S_{20}	1	0.358500	0.012250	21	1	1	20
H_1	0.30	P_2	2	0.142900	0.004050	5	2	2	5
H_1	0.30	P_3	2	0.176900	0.011450	7	2	2	7
H_1	0.30	P_4	2	0.209500	0.003300	9	2	2	9
H_1	0.30	P_5	2	0.244400	0.008050	11	2	2	11
H_1	0.30	P_6	2	0.277500	0.013200	13	2	2	13
H_1	0.30	P_7	1	0.157500	0.008750	8	1	1	7
H_1	0.30	P_8	1	0.173100	0.003250	9	1	1	8
H_1	0.30	P_9	1	0.191100	0.005650	10	1	1	9
H_1	0.30	P_{10}	1	0.206900	0.011300	11	1	1	10

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Table S10 (continued)

Field	δ	Method	Cycles	Median	IQR	F	J	LU	Solves
H_1	0.30	P_{11}	1	0.224700	0.008100	12	1	1	11
H_1	0.30	P_{12}	1	0.241600	0.009050	13	1	1	12
H_1	0.30	P_{13}	1	0.257500	0.009650	14	1	1	13
H_1	0.30	P_{14}	1	0.273900	0.009900	15	1	1	14
H_1	0.30	P_{15}	1	0.294900	0.011500	16	1	1	15
H_1	0.30	P_{16}	1	0.306600	0.010450	17	1	1	16
H_1	0.30	P_{17}	1	0.325400	0.009200	18	1	1	17
H_1	0.30	P_{18}	1	0.342400	0.009600	19	1	1	18
H_1	0.30	P_{19}	1	0.357400	0.008700	20	1	1	19
H_1	0.30	P_{20}	1	0.374700	0.007950	21	1	1	20
H_1	0.30	S_1	3	0.142100	0.003750	4	3	3	3
H_1	0.30	S_2	2	0.136000	0.008450	5	2	2	4
H_1	0.30	S_3	2	0.168900	0.003850	7	2	2	6
H_1	0.30	S_4	2	0.204300	0.009900	9	2	2	8
H_1	0.30	S_5	2	0.235900	0.008300	11	2	2	10
H_1	0.30	S_6	2	0.271000	0.006450	13	2	2	12
H_1	0.30	S_7	1	0.158300	0.004150	8	1	1	7
H_1	0.30	S_8	1	0.175800	0.008050	9	1	1	8
H_1	0.30	S_9	1	0.191800	0.003150	10	1	1	9
H_1	0.30	S_{10}	1	0.208300	0.006650	11	1	1	10
H_1	0.30	S_{11}	1	0.226000	0.013400	12	1	1	11
H_1	0.30	S_{12}	1	0.244000	0.015000	13	1	1	12
H_1	0.30	S_{13}	1	0.259700	0.009550	14	1	1	13
H_1	0.30	S_{14}	1	0.276000	0.010200	15	1	1	14
H_1	0.30	S_{15}	1	0.293400	0.010700	16	1	1	15
H_1	0.30	S_{16}	1	0.308600	0.007650	17	1	1	16
H_1	0.30	S_{17}	1	0.324500	0.012650	18	1	1	17
H_1	0.30	S_{18}	1	0.342200	0.010200	19	1	1	18
H_1	0.30	S_{19}	1	0.362900	0.013850	20	1	1	19
H_1	0.30	S_{20}	1	0.376700	0.010550	21	1	1	20
H_1	0.60	P_2	2	0.138100	0.011950	5	2	2	5
H_1	0.60	P_3	2	0.168600	0.006450	7	2	2	7
H_1	0.60	P_4	2	0.202000	0.009550	9	2	2	9
H_1	0.60	P_5	2	0.232600	0.006600	11	2	2	11
H_1	0.60	P_6	2	0.267000	0.012300	13	2	2	13
H_1	0.60	P_7	2	0.300400	0.011250	15	2	2	15
H_1	0.60	P_8	2	0.331900	0.007650	17	2	2	17
H_1	0.60	P_9	1	0.184200	0.005150	10	1	1	9
H_1	0.60	P_{10}	1	0.196900	0.007550	11	1	1	10
H_1	0.60	P_{11}	1	0.215300	0.006400	12	1	1	11
H_1	0.60	P_{12}	1	0.231200	0.012150	13	1	1	12
H_1	0.60	P_{13}	1	0.249200	0.016750	14	1	1	13
H_1	0.60	P_{14}	1	0.264900	0.010250	15	1	1	14
H_1	0.60	P_{15}	1	0.279500	0.008850	16	1	1	15
H_1	0.60	P_{16}	1	0.297200	0.012100	17	1	1	16
H_1	0.60	P_{17}	1	0.313600	0.025300	18	1	1	17
H_1	0.60	P_{18}	1	0.328700	0.009650	19	1	1	18
H_1	0.60	P_{19}	1	0.344700	0.008650	20	1	1	19
H_1	0.60	P_{20}	1	0.358800	0.009900	21	1	1	20
H_1	0.60	S_1	4	0.176600	0.007950	5	4	4	4
H_1	0.60	S_2	2	0.129300	0.003950	5	2	2	4
H_1	0.60	S_3	2	0.162900	0.005000	7	2	2	6
H_1	0.60	S_4	2	0.194500	0.006800	9	2	2	8
H_1	0.60	S_5	2	0.227400	0.011350	11	2	2	10
H_1	0.60	S_6	2	0.260500	0.007750	13	2	2	12
H_1	0.60	S_7	2	0.292500	0.010700	15	2	2	14
H_1	0.60	S_8	2	0.324600	0.011700	17	2	2	16
H_1	0.60	S_9	1	0.183700	0.008600	10	1	1	9
H_1	0.60	S_{10}	1	0.200600	0.006700	11	1	1	10

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Table S10 (continued)

Field	δ	Method	Cycles	Median	IQR	F	J	LU	Solves
H_1	0.60	S_{11}	1	0.217700	0.009050	12	1	1	11
H_1	0.60	S_{12}	1	0.233300	0.007000	13	1	1	12
H_1	0.60	S_{13}	1	0.250400	0.009150	14	1	1	13
H_1	0.60	S_{14}	1	0.263100	0.007850	15	1	1	14
H_1	0.60	S_{15}	1	0.279700	0.009150	16	1	1	15
H_1	0.60	S_{16}	1	0.297200	0.009350	17	1	1	16
H_1	0.60	S_{17}	1	0.315100	0.013550	18	1	1	17
H_1	0.60	S_{18}	1	0.327100	0.010650	19	1	1	18
H_1	0.60	S_{19}	1	0.346300	0.015700	20	1	1	19
H_1	0.60	S_{20}	1	0.363500	0.020350	21	1	1	20
H_5	0.10	P_2	2	1.878800	0.260300	5	2	2	5
H_5	0.10	P_3	2	1.964800	0.206650	7	2	2	7
H_5	0.10	P_4	1	1.073800	0.137300	5	1	1	4
H_5	0.10	P_5	1	1.123100	0.111150	6	1	1	5
H_5	0.10	P_6	1	1.169500	0.126550	7	1	1	6
H_5	0.10	P_7	1	1.170000	0.153900	8	1	1	7
H_5	0.10	P_8	1	1.296800	0.085600	9	1	1	8
H_5	0.10	P_9	1	1.290400	0.170550	10	1	1	9
H_5	0.10	P_{10}	1	1.422700	0.148700	11	1	1	10
H_5	0.10	P_{11}	1	1.402100	0.177800	12	1	1	11
H_5	0.10	P_{12}	1	1.445400	0.157750	13	1	1	12
H_5	0.10	P_{13}	1	1.506300	0.143900	14	1	1	13
H_5	0.10	P_{14}	1	1.488900	0.112950	15	1	1	14
H_5	0.10	P_{15}	1	1.545700	0.118400	16	1	1	15
H_5	0.10	P_{16}	1	1.652800	0.136150	17	1	1	16
H_5	0.10	P_{17}	1	1.659600	0.157750	18	1	1	17
H_5	0.10	P_{18}	1	1.735700	0.123100	19	1	1	18
H_5	0.10	P_{19}	1	1.783200	0.177850	20	1	1	19
H_5	0.10	P_{20}	1	1.814000	0.221800	21	1	1	20
H_5	0.10	S_1	3	2.570300	0.248200	4	3	3	3
H_5	0.10	S_2	2	1.842400	0.289850	5	2	2	4
H_5	0.10	S_3	2	2.011100	0.224650	7	2	2	6
H_5	0.10	S_4	1	1.054900	0.128150	5	1	1	4
H_5	0.10	S_5	1	1.080800	0.137900	6	1	1	5
H_5	0.10	S_6	1	1.160300	0.116950	7	1	1	6
H_5	0.10	S_7	1	1.174400	0.100350	8	1	1	7
H_5	0.10	S_8	1	1.247800	0.120250	9	1	1	8
H_5	0.10	S_9	1	1.314400	0.152050	10	1	1	9
H_5	0.10	S_{10}	1	1.342800	0.169100	11	1	1	10
H_5	0.10	S_{11}	1	1.381700	0.210600	12	1	1	11
H_5	0.10	S_{12}	1	1.415200	0.181250	13	1	1	12
H_5	0.10	S_{13}	1	1.479000	0.197250	14	1	1	13
H_5	0.10	S_{14}	1	1.523300	0.128950	15	1	1	14
H_5	0.10	S_{15}	1	1.571900	0.140650	16	1	1	15
H_5	0.10	S_{16}	1	1.667300	0.203500	17	1	1	16
H_5	0.10	S_{17}	1	1.636200	0.105700	18	1	1	17
H_5	0.10	S_{18}	1	1.732000	0.194900	19	1	1	18
H_5	0.10	S_{19}	1	1.764600	0.178200	20	1	1	19
H_5	0.10	S_{20}	1	1.809300	0.153700	21	1	1	20
H_5	0.30	P_2	2	1.679000	0.101950	5	2	2	5
H_5	0.30	P_3	2	1.757000	0.076900	7	2	2	7
H_5	0.30	P_4	2	1.881200	0.140850	9	2	2	9
H_5	0.30	P_5	1	0.988600	0.041050	6	1	1	5
H_5	0.30	P_6	1	1.036200	0.099250	7	1	1	6
H_5	0.30	P_7	1	1.098900	0.092750	8	1	1	7
H_5	0.30	P_8	1	1.115700	0.067700	9	1	1	8
H_5	0.30	P_9	1	1.160800	0.088050	10	1	1	9
H_5	0.30	P_{10}	1	1.212000	0.047100	11	1	1	10
H_5	0.30	P_{11}	1	1.272500	0.114850	12	1	1	11

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Table S10 (continued)

Field	δ	Method	Cycles	Median	IQR	F	J	LU	Solves
H_5	0.30	P_{12}	1	1.292800	0.070050	13	1	1	12
H_5	0.30	P_{13}	1	1.371400	0.111950	14	1	1	13
H_5	0.30	P_{14}	1	1.399000	0.037550	15	1	1	14
H_5	0.30	P_{15}	1	1.437500	0.061750	16	1	1	15
H_5	0.30	P_{16}	1	1.480000	0.041700	17	1	1	16
H_5	0.30	P_{17}	1	1.542500	0.069100	18	1	1	17
H_5	0.30	P_{18}	1	1.576200	0.153800	19	1	1	18
H_5	0.30	P_{19}	1	1.612500	0.096900	20	1	1	19
H_5	0.30	P_{20}	1	1.664000	0.056650	21	1	1	20
H_5	0.30	S_1	3	2.327600	0.198300	4	3	3	3
H_5	0.30	S_2	2	1.654800	0.159400	5	2	2	4
H_5	0.30	S_3	2	1.730000	0.088750	7	2	2	6
H_5	0.30	S_4	2	1.856100	0.118000	9	2	2	8
H_5	0.30	S_5	1	0.999200	0.067750	6	1	1	5
H_5	0.30	S_6	1	1.039600	0.056500	7	1	1	6
H_5	0.30	S_7	1	1.096200	0.059850	8	1	1	7
H_5	0.30	S_8	1	1.135100	0.070800	9	1	1	8
H_5	0.30	S_9	1	1.171600	0.087100	10	1	1	9
H_5	0.30	S_{10}	1	1.226800	0.087200	11	1	1	10
H_5	0.30	S_{11}	1	1.262500	0.052900	12	1	1	11
H_5	0.30	S_{12}	1	1.338400	0.095050	13	1	1	12
H_5	0.30	S_{13}	1	1.338400	0.047150	14	1	1	13
H_5	0.30	S_{14}	1	1.406200	0.119400	15	1	1	14
H_5	0.30	S_{15}	1	1.456900	0.101750	16	1	1	15
H_5	0.30	S_{16}	1	1.505700	0.128650	17	1	1	16
H_5	0.30	S_{17}	1	1.535800	0.073050	18	1	1	17
H_5	0.30	S_{18}	1	1.603100	0.107250	19	1	1	18
H_5	0.30	S_{19}	1	1.634600	0.080500	20	1	1	19
H_5	0.30	S_{20}	1	1.675000	0.059700	21	1	1	20
H_5	0.60	P_2	2	1.809900	0.174800	5	2	2	5
H_5	0.60	P_3	2	1.870900	0.220450	7	2	2	7
H_5	0.60	P_4	2	1.973900	0.194500	9	2	2	9
H_5	0.60	P_5	2	2.068100	0.237400	11	2	2	11
H_5	0.60	P_6	1	1.056200	0.119100	7	1	1	6
H_5	0.60	P_7	1	1.121500	0.130400	8	1	1	7
H_5	0.60	P_8	1	1.168400	0.149750	9	1	1	8
H_5	0.60	P_9	1	1.203700	0.127550	10	1	1	9
H_5	0.60	P_{10}	1	1.284600	0.123000	11	1	1	10
H_5	0.60	P_{11}	1	1.303000	0.131400	12	1	1	11
H_5	0.60	P_{12}	1	1.391300	0.142950	13	1	1	12
H_5	0.60	P_{13}	1	1.397400	0.203900	14	1	1	13
H_5	0.60	P_{14}	1	1.439400	0.142550	15	1	1	14
H_5	0.60	P_{15}	1	1.519000	0.130750	16	1	1	15
H_5	0.60	P_{16}	1	1.530200	0.154600	17	1	1	16
H_5	0.60	P_{17}	1	1.613800	0.164600	18	1	1	17
H_5	0.60	P_{18}	1	1.623500	0.154000	19	1	1	18
H_5	0.60	P_{19}	1	1.666400	0.179750	20	1	1	19
H_5	0.60	P_{20}	1	1.717400	0.100500	21	1	1	20
H_5	0.60	S_1	3	2.393800	0.169400	4	3	3	3
H_5	0.60	S_2	2	1.693200	0.279600	5	2	2	4
H_5	0.60	S_3	2	1.857500	0.278600	7	2	2	6
H_5	0.60	S_4	2	1.902800	0.244950	9	2	2	8
H_5	0.60	S_5	2	2.009600	0.140500	11	2	2	10
H_5	0.60	S_6	1	1.074300	0.133500	7	1	1	6
H_5	0.60	S_7	1	1.133700	0.174500	8	1	1	7
H_5	0.60	S_8	1	1.154100	0.171150	9	1	1	8
H_5	0.60	S_9	1	1.259900	0.149400	10	1	1	9
H_5	0.60	S_{10}	1	1.253300	0.111700	11	1	1	10
H_5	0.60	S_{11}	1	1.304500	0.144950	12	1	1	11

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Table S10 (continued)

Field	δ	Method	Cycles	Median	IQR	F	J	LU	Solves
H_5	0.60	S_{12}	1	1.368100	0.133400	13	1	1	12
H_5	0.60	S_{13}	1	1.421400	0.187300	14	1	1	13
H_5	0.60	S_{14}	1	1.472200	0.197850	15	1	1	14
H_5	0.60	S_{15}	1	1.471400	0.097000	16	1	1	15
H_5	0.60	S_{16}	1	1.516200	0.158250	17	1	1	16
H_5	0.60	S_{17}	1	1.601300	0.121000	18	1	1	17
H_5	0.60	S_{18}	1	1.644800	0.183350	19	1	1	18
H_5	0.60	S_{19}	1	1.667100	0.137100	20	1	1	19
H_5	0.60	S_{20}	1	1.732800	0.082500	21	1	1	20

The I/D summaries are retained at their original printed precision in Table S11; they must not be mistaken for recovered full-precision repetition samples.

Table S11: Immediate (I) versus delayed (D) initialization. Milliseconds; medians and IQRs as rounded in the article. The individual repetition samples are not available in the archived export.

Field	Method	Cycles	I/D	I time (IQR)	D time (IQR)	Δt_D [%]
H_1	P_3	1/2		0.1097 (0.0018)	0.1747 (0.0080)	+59.3
H_5	P_9	1/1		1.8518 (0.1072)	1.1357 (0.0424)	-38.7
H_1	P_5	1/1		0.1416 (0.0103)	0.1172 (0.0045)	-17.2
H_2	P_5	1/1		0.2102 (0.0143)	0.1640 (0.0103)	-22.0
H_3	P_5	1/1		0.3216 (0.0173)	0.2281 (0.0167)	-29.1
H_4	P_5	1/1		0.7934 (0.0661)	0.5076 (0.0313)	-36.0
H_5	P_5	1/1		1.9275 (0.1812)	1.1219 (0.1758)	-41.8
H_1	P_7	1/1		0.1744 (0.0062)	0.1490 (0.0020)	-14.6
H_2	P_7	1/1		0.2475 (0.0105)	0.2011 (0.0113)	-18.7
H_3	P_7	1/1		0.3629 (0.0126)	0.2745 (0.0109)	-24.4
H_4	P_7	1/1		0.8560 (0.0569)	0.5802 (0.0263)	-32.2
H_5	P_7	1/1		2.0143 (0.2427)	1.2053 (0.1706)	-40.2

8 Full sensitivity breakdown and auxiliary plots

The following table retains all six control points for the stationary optima and near-optimal bands. Interval stability is established by the affine normalized-cost arguments in the article, not inferred from this finite list alone.

Table S12: Stationary family optima and 0.5% near-optimal bands.

Field	κ	N_P^*	m_S^*	$10^6 \eta_P^*$	$10^6 \eta_S^*$	Advantage [%]	P/S bands
H_1	1	3	3	172.461715	165.035043	4.50	{3}/{3}
H_1	1.2	3	3	171.409882	164.058504	4.48	{3}/{3}
H_1	1.298	3	3	170.899153	163.584207	4.47	{3}/{3}
H_1	1.7	3	3	168.835587	161.666981	4.43	{3}/{3}
H_1	3.23511	3	3	161.393776	154.741466	4.30	{3}/{3}
H_1	4.45444	3	3	155.934456	149.649474	4.20	{3}/{3}
H_5	1	9	10	0.555516	0.543469	2.22	{8, 9, 10, 11}/{9, 10, 11}
H_5	1.2	9	10	0.554944	0.542925	2.21	{8, 9, 10, 11}/{9, 10, 11}
H_5	1.298	9	10	0.554664	0.542659	2.21	{8, 9, 10, 11}/{9, 10, 11}
H_5	1.7	9	10	0.553518	0.541570	2.21	{8, 9, 10, 11}/{9, 10, 11}
H_5	3.23511	9	10	0.549186	0.537452	2.18	{8, 9, 10, 11}/{9, 10, 11, 12}
H_5	4.45444	9	10	0.545794	0.534226	2.17	{8, 9, 10, 11}/{9, 10, 11, 12}

Figures S1 and S2 are additional views of the same archived timing summaries. The efficiency-region figures in the article answer a different question: how the stationary favorable set varies with (μ_0, μ_1) and dimension.

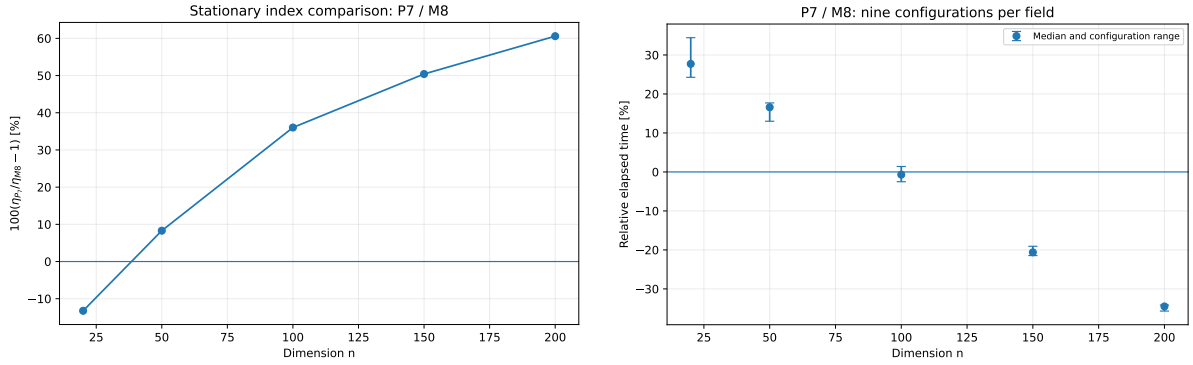


Figure S1: P_7/M_8 : stationary index advantage (left) and median with configuration range of relative elapsed time (right). Positive index differences and negative time differences favor P_7 . The displayed range is not an IQR or confidence interval.

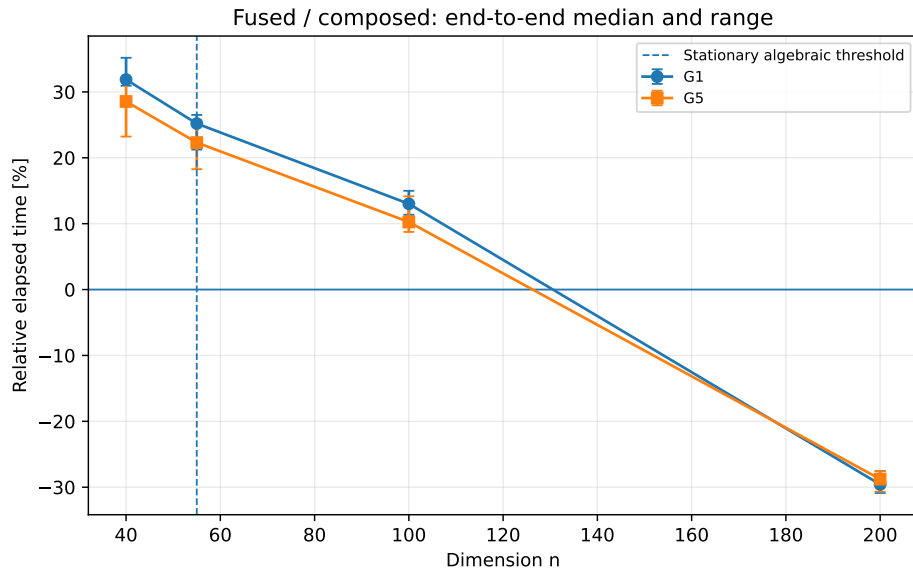


Figure S2: End-to-end $\hat{P}_2/(P_2 \circ P_2)$: median and range of relative elapsed time. The line $n = 55$ marks the stationary algebraic threshold, not an estimated temporal crossover. No interpolation establishes universal monotonicity.

9 Multiprecision comparison of independently selected family optima

All eight groups in the frozen family plan are complete. Table S13 states the coverage. The dataset contains 496 individual timed full solves, or 248 paired repetitions, across 16 method configurations. The four H1 groups share one recorded session; each H5 tolerance has a separate session, for five sessions in total. Within each group, both methods were interleaved in the same session. No pilot sample, partial group or across-session splice is included.

The comparison remains P_3/S_3 in H_1 ($n = 20$) and P_9/S_{10} in H_5 ($n = 200$), the independent optima of the canonical reference-index model at $\kappa = 1$. These are not calibrated multiprecision time optima. The central distance $\delta = 0.30$ and all four initially proposed tolerances were retained in this resource-limited tranche. The plan was frozen after feasibility pilots and before repeated timing runs. Pilot outcomes were visible at the design decision; blinded selection is not claimed.

Table S13: Complete frozen family plan. Every group contains 31 repetitions of each method.

Field	Pair	TOL	Status
H_1	P_3/S_3	10^{-50}	Complete
H_1	P_3/S_3	10^{-200}	Complete
H_1	P_3/S_3	10^{-800}	Complete
H_1	P_3/S_3	10^{-1024}	Complete
H_5	P_9/S_{10}	10^{-50}	Complete
H_5	P_9/S_{10}	10^{-200}	Complete
H_5	P_9/S_{10}	10^{-800}	Complete
H_5	P_9/S_{10}	10^{-1024}	Complete

9.1 Execution identity, arithmetic and measured boundaries

The source and protocol hashes identify the actual frozen files, not a subsequent public release. The source hash is `f62dd86b1c521d14e1f6e5bdaf9132482767b5be157758c29b9e2ae77f3dc66d` and the protocol hash is `7eddefb2878492a4ed2d4d37db8f6b3d74de79f342298462b8e5e8950e489def`. The original driver hashes file bytes indexed by platform-native relative paths; the audit reconstructs Windows separators when checking this identity on another platform. Program bytes are unchanged.

All five session records agree on Windows-11-10.0.26200-SP0; AMD64 Family 25 Model 80 Stepping 0, AuthenticAMD; Python 3.13.6; NumPy 2.3.2; mpmath 1.3.0, Python backend; and no gmpy2. The solver is serial, uses mpmath numbers in object arrays and does not call BLAS. A generic partial-pivoting LU is computed in multiprecision and reused by triangular solves; no inverse or special low-rank representation is formed. Six coefficient matrices (A/B/C at dimensions 20 and 200) agree bitwise with the prescribed PCG64 realization.

The float64 coefficient matrices are lifted as exact dyadic data. Decimal analytic constants and $\alpha_j = j/10$ are evaluated at the current precision. One initial vector constructed at 1300 digits is frozen and lifted exactly for the 2600-digit check. This preserves the coefficient realization but does not reproduce float64 evaluation roundoff. Original x coordinates are iterated. Problem generation and constant matrix prescaling are outside the timer.

The settings are 1300 working digits, 2600 verification digits, delayed D initialization, 20 maximum cycles, two warmups and 31 repetitions. Warmups are specified in the frozen plan and driver but not saved as separate timed observations. The timer is `time.perf_counter_ns`, using the recorded monotonic QueryPerformanceCounter clock. It measures the complete solve, including copies, counters, numerical operations and residual tests. Error norms, trace

serialization, I/O, precision gates and statistical aggregation are outside that interval. Integer nanosecond readings do not imply nanosecond experimental accuracy.

Every repetition has one measurement of each method in deterministic pseudo-random order. Exact order, durations, status, resources and final error/residual strings are retained under `data/multiprecision/families/sessions`. Each completed-group marker identifies its session. For H5 the session starts are September 26, 2026, at 04:35, 16:20 and 19:13 UTC for $d = 50, 200, 800$, and September 27 at 12:17 UTC for $d = 1024$. The H1 session starts on September 26 at 04:31 UTC. These identify recorded invocations, not independent replications of the hardware experiment.

9.2 Complete timing and work summaries

All eight groups, favorable or otherwise, appear below. Times are seconds, unlike the earlier millisecond tables. Reference work uses the unchanged model, not physical scalar-operation weights calibrated for multiprecision.

Table S14: All eight family comparisons: P_3/S_3 in H_1 and P_9/S_{10} in H_5 . Times are seconds with IQRs in parentheses; d denotes $\text{TOL} = 10^{-d}$. Negative differences favor P . The filter is the two-IQR rule.

Field	d : $\text{TOL} = 10^{-d}$	cycles P/S	$\tilde{t}_P$ (IQR)	$\tilde{t}_S$ (IQR)	Δt [%]	ΔW [%]	filter
H_1	50	3/3	0.357824 (0.138745)	0.355855 (0.163756)	+0.553	+3.026	no
H_1	200	4/4	0.531415 (0.058401)	0.502425 (0.089211)	+5.770	+3.444	no
H_1	800	5/5	0.537007 (0.009490)	0.528849 (0.011225)	+1.543	+3.700	no
H_1	1024	5/5	0.530157 (0.006841)	0.525820 (0.007198)	+0.825	+3.700	no
H_5	50	2/2	134.308598 (0.385663)	138.014224 (0.347079)	-2.685	-3.267	yes
H_5	200	2/2	136.036516 (3.976849)	140.439827 (4.375426)	-3.135	-3.267	no
H_5	800	3/3	190.385437 (2.253746)	192.323630 (1.502451)	-1.008	-3.133	no
H_5	1024	3/3	188.605813 (0.335320)	191.176665 (1.062469)	-1.345	-3.133	no

The H1 medians favor S3 in all four configurations, by 0.553% to 5.770% relative to S3, without crossing the descriptive two-IQR filter. At 10^{-200} P3 produces a much smaller error without saving a cycle. The additional predictor solves increase its reference work by 3.026% to 3.700%.

In H5, P9 has lower medians at all four tolerances: reductions of 2.685%, 3.135%, 1.008% and 1.345% in increasing d order. Only the 10^{-50} difference passes the two-IQR filter. For 10^{-1024} , the difference of medians is 2.5708524 s, below the 2.7955777 s threshold. P9 and S10 use two cycles for $d = 50, 200$ and three for $d = 800, 1024$; P9 activates inherited memory in one or two cycles, respectively. Its reference-work reduction is 3.267% for the two-cycle runs and 3.133% for the three-cycle runs. No family comparison saves an outer cycle. Because correction counts also differ, the H5 outcome supports those optimized pairs in their recorded sessions, not a causal isolation of memory.

Table S15: Exact total resources, including the terminal residual evaluation. Here d denotes $\text{TOL} = 10^{-d}$. Additional matrix–vector products and comparator scalings are zero for these family runs.

Field	d	Method	cycles	active memory	F	J / LU	solves	work
H_1	50	P_3	3	2	10	3 / 3	11	27240
H_1	50	S_3	3	0	10	3 / 3	9	26440
H_1	200	P_3	4	3	13	4 / 4	15	36040
H_1	200	S_3	4	0	13	4 / 4	12	34840
H_1	800	P_3	5	4	16	5 / 5	19	44840
H_1	800	S_3	5	0	16	5 / 5	15	43240
H_1	1024	P_3	5	4	16	5 / 5	19	44840
H_1	1024	S_3	5	0	16	5 / 5	15	43240
H_5	50	P_9	2	1	19	2 / 2	19	8658200
H_5	50	S_{10}	2	0	21	2 / 2	20	8950600
H_5	200	P_9	2	1	19	2 / 2	19	8658200
H_5	200	S_{10}	2	0	21	2 / 2	20	8950600
H_5	800	P_9	3	2	28	3 / 3	29	12944200
H_5	800	S_{10}	3	0	31	3 / 3	30	13362800
H_5	1024	P_9	3	2	28	3 / 3	29	12944200
H_5	1024	S_{10}	3	0	31	3 / 3	30	13362800

9.3 Paired bootstrap and descriptive separation

For each of 10000 bootstrap resamples, 31 pair indices are drawn with replacement and the same indices are used for both methods. The statistic is the ratio of the resampled medians, not the median of within-pair ratios. The 2.5th and 97.5th percentiles give Table S16; the deterministic seed follows from the frozen driver, plan and group ID.

Table S16: Ratio of medians and pointwise paired percentile bootstrap intervals (nominal 95%). No multiplicity or serial-dependence correction is applied.

Field	d	ratio P/S	lower	upper	two-IQR filter
H_1	50	1.005532	0.889939	1.075961	no
H_1	200	1.057699	0.976482	1.088468	no
H_1	800	1.015425	1.009291	1.020359	no
H_1	1024	1.008248	1.003821	1.014380	no
H_5	50	0.973150	0.972079	0.974394	yes
H_5	200	0.968646	0.957478	0.977008	no
H_5	800	0.989922	0.987533	0.994525	no
H_5	1024	0.986552	0.984817	0.987467	no

The criteria are not interchangeable. The H_1 intervals at 10^{-800} and 10^{-1024} lie above one, although the two-IQR rule is not met; at 10^{-50} and 10^{-200} they include one. All four H_5 intervals lie below one, but only its 10^{-50} group passes the two-IQR filter. Failure to pass the latter is not evidence of equal times. These summaries are conditional on the recorded sessions and sampling design; they do not establish universal superiority or a quantitative overhead model. No historical IQR-only table is bootstrapped.

9.4 Precision checks, termination and saved trajectories

Each method was checked before its repeated timings at 1300 and 2600 digits, using the same frozen start. The final low-precision iterate was also reevaluated at 2600 digits. All sixteen checks pass and cycle counts agree. The audit recalculates error norms and COC from the saved exact dyadics and independently evaluates the final residuals by row-wise sums of the stated field formula. These are numerical checks, not interval certification.

The comparison uses the floor and criterion

$$f = 10^{-1300+40}(1 + \|\alpha\|_2), \quad \frac{\|x_k^{(1300)} - x_k^{(2600)}\|_2}{\max\{\|x_k^{(2600)} - \alpha\|_2, f\}} < 10^{-10}.$$

A high-precision error below f triggers a floor-censored flag; passing the test then is not a relative-accuracy certificate. Separately, the reevaluated low-precision final iterate must meet the requested residual tolerance.

Table S17: Final 1300-digit iterate reevaluated at 2600 digits. The floor column denotes a censored precision comparison, not failure to meet the tolerance.

Field	d	Method	checked residual	checked error	final floor
H_1	50	P_3	1.375×10^{-153}	3.396×10^{-154}	no
H_1	50	S_3	1.785×10^{-120}	4.526×10^{-121}	no
H_1	200	P_3	7.705×10^{-705}	1.956×10^{-705}	no
H_1	200	S_3	2.284×10^{-485}	5.385×10^{-486}	no
H_1	800	P_3	6.733×10^{-1301}	1.697×10^{-1301}	yes
H_1	800	S_3	6.733×10^{-1301}	1.697×10^{-1301}	yes
H_1	1024	P_3	6.733×10^{-1301}	1.697×10^{-1301}	yes
H_1	1024	S_3	6.733×10^{-1301}	1.697×10^{-1301}	yes
H_5	50	P_9	2.637×10^{-286}	6.679×10^{-287}	no
H_5	50	S_{10}	7.598×10^{-321}	1.941×10^{-321}	no
H_5	200	P_9	2.637×10^{-286}	6.679×10^{-287}	no
H_5	200	S_{10}	7.598×10^{-321}	1.941×10^{-321}	no
H_5	800	P_9	2.536×10^{-1299}	6.371×10^{-1300}	yes
H_5	800	S_{10}	2.536×10^{-1299}	6.371×10^{-1300}	yes
H_5	1024	P_9	2.536×10^{-1299}	6.371×10^{-1300}	yes
H_5	1024	S_{10}	2.536×10^{-1299}	6.371×10^{-1300}	yes

For H_1 at 10^{-800} and 10^{-1024} , the final computed error and residual are zero, but the saved iterate has a reevaluated residual of 6.7330×10^{-1301} . For H_5 at both stringent tolerances, the reevaluated residual is 2.5362×10^{-1299} , again below the requested tolerance despite computed zero finals. No COC is formed with these zeros. In H_5 , the only nonzero-error COC uses the initial transient; its third-cycle final is zero at both working precisions. Thus this H_5 campaign supplies no wholly post-startup COC. The established order-verification study remains separate.

Eight nominal paired configurations correspond to five paired exact paths in two fields. Equality was checked directly on every stored dyadic iterate, at both precisions, within each repeated-path pair: H_1 at $d = 800/1024$, H_5 at $d = 50/200$, and H_5 at $d = 800/1024$. These duplicated paths are not additional nonlinear problems. In particular, the H_5 differences in median and IQR between $d = 50$ and $d = 200$, or between $d = 800$ and $d = 1024$, are not attributed to an algorithmic path change; they were timed in different sessions. All measured outcomes are retained.

Table S18 displays the longest saved path in each field at both precisions. Every per-group trace remains in `data/multiprecision/families/audit/stored_trace_summary.csv`, and the exact vectors are in the compressed gate files. A dash denotes an unavailable COC, not zero. The shorter-tolerance paths are prefixes, checked against these saved vectors.

Table S18: Saved trajectory diagnostics at 10^{-1024} for both fields. Shorter-tolerance runs follow prefixes of these paths. “Tail” requires a defined COC wholly after the initial block; a zero final is not used for COC.

Field	Method	digits	k	error	residual	COC	tail
H_1	P_3	1300	0	3.000×10^{-1}	1.233×10^0	–	no
H_1	P_3	1300	1	5.489×10^{-7}	2.286×10^{-6}	–	no
H_1	P_3	1300	2	2.434×10^{-33}	9.356×10^{-33}	4.593070	no
H_1	P_3	1300	3	3.396×10^{-154}	1.375×10^{-153}	4.585969	yes
H_1	P_3	1300	4	1.956×10^{-705}	7.705×10^{-705}	4.561155	yes
H_1	P_3	1300	5	0	0	–	no

Table S18 continued

Field	Method	digits	k	error	residual	COC	tail
H_1	P_3	2600	0	3.000×10^{-1}	1.233×10^0	–	no
H_1	P_3	2600	1	5.489×10^{-7}	2.286×10^{-6}	–	no
H_1	P_3	2600	2	2.434×10^{-33}	9.356×10^{-33}	4.593070	no
H_1	P_3	2600	3	3.396×10^{-154}	1.375×10^{-153}	4.585969	yes
H_1	P_3	2600	4	1.956×10^{-705}	7.705×10^{-705}	4.561155	yes
H_1	P_3	2600	5	0	0	–	no
H_1	S_3	1300	0	3.000×10^{-1}	1.233×10^0	–	no
H_1	S_3	1300	1	5.489×10^{-7}	2.286×10^{-6}	–	no
H_1	S_3	1300	2	8.761×10^{-30}	3.620×10^{-29}	3.973242	no
H_1	S_3	1300	3	4.526×10^{-121}	1.785×10^{-120}	4.004346	yes
H_1	S_3	1300	4	5.385×10^{-486}	2.284×10^{-485}	3.997560	yes
H_1	S_3	1300	5	0	0	–	no
H_1	S_3	2600	0	3.000×10^{-1}	1.233×10^0	–	no
H_1	S_3	2600	1	5.489×10^{-7}	2.286×10^{-6}	–	no
H_1	S_3	2600	2	8.761×10^{-30}	3.620×10^{-29}	3.973242	no
H_1	S_3	2600	3	4.526×10^{-121}	1.785×10^{-120}	4.004346	yes
H_1	S_3	2600	4	5.385×10^{-486}	2.284×10^{-485}	3.997560	yes
H_1	S_3	2600	5	1.231×10^{-1945}	5.098×10^{-1945}	3.999843	yes
H_5	P_9	1300	0	3.000×10^{-1}	1.223×10^0	–	no
H_5	P_9	1300	1	2.752×10^{-25}	1.076×10^{-24}	–	no
H_5	P_9	1300	2	6.679×10^{-287}	2.637×10^{-286}	10.883595	no
H_5	P_9	1300	3	0	0	–	no
H_5	P_9	2600	0	3.000×10^{-1}	1.223×10^0	–	no
H_5	P_9	2600	1	2.752×10^{-25}	1.076×10^{-24}	–	no
H_5	P_9	2600	2	6.679×10^{-287}	2.637×10^{-286}	10.883595	no
H_5	P_9	2600	3	0	0	–	no
H_5	S_{10}	1300	0	3.000×10^{-1}	1.223×10^0	–	no
H_5	S_{10}	1300	1	6.208×10^{-28}	2.414×10^{-27}	–	no
H_5	S_{10}	1300	2	1.941×10^{-321}	7.598×10^{-321}	10.999222	no
H_5	S_{10}	1300	3	0	0	–	no
H_5	S_{10}	2600	0	3.000×10^{-1}	1.223×10^0	–	no
H_5	S_{10}	2600	1	6.208×10^{-28}	2.414×10^{-27}	–	no
H_5	S_{10}	2600	2	1.941×10^{-321}	7.598×10^{-321}	10.999222	no
H_5	S_{10}	2600	3	0	0	–	no

9.5 Completed scope and provenance

All eight planned family groups have completion markers, with 496 raw timing observations and sixteen precision gates. No value is borrowed from a pilot or inferred from cycle counts. The multiprecision timing evidence concerns only the stated optimum-family comparisons at $\delta = 0.30$. This section concerns only the family comparison. External multiprecision measurements are reported separately below; the preceding fused- q timing tables still concern float64 measurements.

Original files are preserved bitwise under `data/multiprecision/families`; reconstructed summaries and the numerical audit are under `data/multiprecision/families/audit`; the exact solver snapshot is under `experiments/multiprecision/families`. The code, frozen protocol and previously supplied five groups remain unchanged. The multiprecision data and expanded supplement are distributed with the repository and are not claimed to belong to Zenodo v1.0.0.

10 External multiprecision comparisons with platform-resolved allocation

The external campaign contains all 16 planned groups: $P_5/M6$ and $P_7/M8$, H_1 ($n = 20$) and H_5 ($n = 200$), $\delta = 0.30$, and residual tolerances 10^{-d} for $d = 50, 200, 800, 1024$. It is separate from the family-optimum campaign and from the earlier float64 external timings. All methods use the stated field, exact-dyadic coefficient lift, original x coordinates, 1300 working digits, 2600 verification digits, partial-pivoting LU reuse and delayed initialization D for P_N .

10.1 Allocation, platform provenance and measurement boundaries

Each group starts with two warmups and 15 paired repetitions. The predeclared rule keeps 15 only when the paired-bootstrap percentile interval excludes one, the two-IQR filter is met, and both favor the same method. Otherwise 16 additional pairs are collected, preceded by new gates and warmups. The initial fixed-size data and decisions are unchanged byte for byte. Eight groups were extended and eight stopped at 15: 736 timed solves, 368 pairs, 32 method configurations and 48 stored precision gates. Warmups are documented by the frozen code and plan, not separately saved as timing observations.

Twelve groups (all eight $P_5/M6$ groups and the four $P_7/M8$ groups in H_1) use the AMD environment. The four $P_7/M8$ groups in H_5 use Intel, each with 15 pairs and no extension. All eight extensions in the current aggregate use AMD, matching their initial phases. There are 16 initial sessions and eight extension sessions; 616 current measurements are AMD and 120 are Intel. No current group pools hardware platforms. Matching platform metadata does not make separate sessions identically distributed or independent.

Table S19: Recorded platforms of the external campaign. “AMD64” in a machine-architecture field is not used to identify the processor vendor.

	AMD	Intel
Processor	AMD64 Family 25 Model 80 Stepping 0, AuthenticAMD	Intel64 Family 6 Model 151 Stepping 2, GenuineIntel
Python	3.13.6	3.13.5
Logical CPUs reported	16	20
OS	Windows-11-10.0.26200-SP0	Windows-11-10.0.26200-SP0
NumPy / mpmath	2.3.2 / 1.3.0	2.3.2 / 1.3.0
Arithmetic	Python backend; no gmpy2	Python backend; no gmpy2
Current groups	12	4

The solver is serial and does not use BLAS. The monotonic `perf_counter_ns` timer wraps the full solve, including copies, counters, scalar/matrix operations and terminal residual tests; it excludes problem setup, error-norm reporting, trace serialization, I/O, gates, warmups and aggregation. Every pair contains one run of each method in a reproducible randomized order. The trial cap is 20 cycles. Full session identifiers and recorded settings are in `data/multiprecision/external/audit/platform_sessions.csv`.

The repeated protocol was frozen on September 27, 2026, before initial timings; allocation decisions were recorded on September 29 after those 16 groups. Eight earlier follow-ups used Intel. They are preserved separately; replacement AMD follow-ups ran on September 30–October 1 using the same eight decisions and no newly selected tolerances. Earlier follow-up outcomes were already visible, so blinded replacement is not claimed. The four unextended $P_7/M8$ groups in H_5 remain Intel; they must not be relabeled as AMD or interpreted as an AMD dimension sweep.

10.2 Final descriptive timing summaries

Table S20: External $P_5/M6$ timings. Seconds; IQR in parentheses. Platforms are identified per row. Negative changes favor P_5 . The adaptive aggregate combines only the same reported platform, but extended groups span two sessions.

Field	platform	d	pairs	cycles	P_5 median (IQR)	$M6$ median (IQR)	change [%]	filter
H_1	AMD	50	31	2/2	0.312290 (0.007706)	0.302739 (0.005472)	+3.155	no
H_1	AMD	200	31	3/3	0.462004 (0.008549)	0.447156 (0.008603)	+3.321	no
H_1	AMD	800	31	4/4	0.553661 (0.008682)	0.572815 (0.008094)	-3.344	no
H_1	AMD	1024	31	4/4	0.556572 (0.010900)	0.571583 (0.012388)	-2.626	no
H_5	AMD	50	15	2/2	114.474034 (0.426745)	112.138516 (0.160686)	+2.083	yes
H_5	AMD	200	31	3/3	173.749752 (2.623016)	168.895049 (2.464180)	+2.874	no
H_5	AMD	800	31	4/4	228.295552 (9.249476)	227.487334 (8.264110)	+0.355	no
H_5	AMD	1024	31	4/4	222.429909 (2.535950)	221.004646 (2.961342)	+0.645	no

Table S21: External $P_7/M8$ timings. Seconds; IQR in parentheses. Platforms are identified per row. Negative changes favor P_7 . The adaptive aggregate combines only the same reported platform, but extended groups span two sessions.

Field	platform	d	pairs	cycles	P_7 median (IQR)	$M8$ median (IQR)	change [%]	filter
H_1	AMD	50	31	2/2	0.385014 (0.009912)	0.361616 (0.008927)	+6.470	no
H_1	AMD	200	15	3/3	0.585748 (0.007271)	0.537445 (0.009296)	+8.988	yes
H_1	AMD	800	15	3/3	0.579601 (0.005976)	0.536758 (0.007237)	+7.982	yes
H_1	AMD	1024	15	3/4	0.580049 (0.003926)	0.680062 (0.003980)	-14.706	yes
H_5	Intel	50	15	2/2	100.705191 (0.402399)	161.158137 (0.502481)	-37.512	yes
H_5	Intel	200	15	3/3	148.041299 (0.340894)	241.791999 (0.263207)	-38.773	yes
H_5	Intel	800	15	3/3	147.898461 (0.562725)	241.762238 (0.628141)	-38.825	yes
H_5	Intel	1024	15	3/3	148.493841 (0.610147)	241.748945 (0.690050)	-38.575	yes

For $P_5/M6$, both methods require 2, 3, 4 and 4 cycles as d increases, in both fields. In H_1 , P_5 has higher medians at $d = 50, 200$ and lower medians at 800, 1024, with no two-IQR separation. In H_5 its medians are always higher; only $d = 50$ passes the filter, favoring $M6$. Reference work is higher for P_5 in all eight groups. Neither this model mismatch nor an unpassed filter is used to assert equal execution times.

For $P_7/M8$ in H_1 , the first three medians favor $M8$; the $d = 200, 800$ differences pass the filter. At $d = 1024$, P_7 takes three cycles against four, lowers the median by 14.706% and lowers reference work by 10.252%. In H_5 (Intel), both methods take 2, 3, 3 and 3 cycles; P_7 lowers the four medians by 37.512–38.825%, and all four differences pass the filter. Its reference-work reduction is about 34.94%. The comparison is within the recorded Intel platform; comparing these reductions to H_1 does not isolate dimension from hardware.

The predictor is active in every P_5/P_7 group. A saved cycle occurs only for $P_7/M8$ in H_1 at $d = 1024$. The experiment compares complete algorithms, not an intervention that isolates the predictor while fixing all other evaluation and factorization counts.

10.3 Session phases and bootstrap limitations

Table S22: Extended groups: absolute median wall times by AMD session phase (seconds). Each initial phase has 15 pairs and each follow-up has 16. The ratios have the same direction in the two phases for all eight groups; their magnitudes and dispersions need not agree.

Pair	field	d	initial A	initial B	follow-up A	follow-up B	initial ratio	follow-up ratio
$P_5/M6$	H1	50	0.307206	0.299413	0.313455	0.304820	1.026028	1.028329
$P_5/M6$	H1	200	0.459218	0.444463	0.465256	0.450878	1.033197	1.031889
$P_5/M6$	H1	800	0.552856	0.573300	0.554224	0.570312	0.964340	0.971790
$P_5/M6$	H1	1024	0.552425	0.568990	0.560870	0.577600	0.970887	0.971035
$P_7/M8$	H1	50	0.383449	0.361157	0.386268	0.364850	1.061722	1.058703
$P_5/M6$	H5	200	176.126698	171.044699	173.685327	168.858873	1.029712	1.028583
$P_5/M6$	H5	800	229.406337	229.357006	222.010974	221.114813	1.000215	1.004053
$P_5/M6$	H5	1024	224.263011	223.337591	221.592648	220.372374	1.004144	1.005537

The current AMD follow-ups retain the direction of the initial ratio in all eight extended groups. Their medians and IQRs need not be stationary across dates. The 31-pair aggregate is a descriptive mixture of two same-platform session samples with fixed sizes 15 and 16, not a single uninterrupted 31-pair experiment. All eight extended aggregates fail the two-IQR filter; additional observations are not guaranteed to produce separation.

Table S23: Paired-bootstrap percentile diagnostics (2.5th and 97.5th percentiles, 10000 resamples): initial fixed-size sample, final adaptive aggregate, and phase-stratified sensitivity. The latter retains 15/16 pairs per phase. None is claimed to have 95% coverage after adaptive allocation; a dash means no extension.

Pair	field	d	final pairs	initial interval	final interval	stratified interval
$P_5/M6$	H1	50	31	[1.022911, 1.042705]	[1.016530, 1.036571]	[1.019026, 1.036571]
$P_5/M6$	H1	200	31	[1.020311, 1.039386]	[1.023216, 1.040560]	[1.023216, 1.040057]
$P_5/M6$	H1	800	31	[0.956404, 0.973812]	[0.958348, 0.974457]	[0.958540, 0.974457]
$P_5/M6$	H1	1024	31	[0.963829, 0.976647]	[0.965758, 0.980707]	[0.966110, 0.980707]
$P_7/M8$	H1	50	31	[1.053063, 1.074850]	[1.051398, 1.073916]	[1.051423, 1.073916]
$P_7/M8$	H1	200	15	[1.078388, 1.100528]	[1.078388, 1.100528]	-
$P_7/M8$	H1	800	15	[1.074540, 1.089658]	[1.074540, 1.089391]	-
$P_7/M8$	H1	1024	15	[0.848738, 0.857983]	[0.848738, 0.857983]	-
$P_5/M6$	H5	50	15	[1.019644, 1.022927]	[1.019727, 1.023259]	-
$P_5/M6$	H5	200	31	[1.027951, 1.040488]	[1.026264, 1.032821]	[1.026264, 1.031870]
$P_5/M6$	H5	800	31	[0.990781, 1.011756]	[0.998501, 1.023605]	[0.998904, 1.011997]
$P_5/M6$	H5	1024	31	[1.000931, 1.006869]	[1.001457, 1.007839]	[1.003710, 1.007878]
$P_7/M8$	H5	50	15	[0.623243, 0.626631]	[0.623243, 0.626631]	-
$P_7/M8$	H5	200	15	[0.611004, 0.612671]	[0.611004, 0.612671]	-
$P_7/M8$	H5	800	15	[0.610620, 0.613952]	[0.610635, 0.613923]	-
$P_7/M8$	H5	1024	15	[0.612782, 0.615891]	[0.613007, 0.615891]	-

The archived runner generates 10000 paired bootstrap resamples and reports the ratio of resampled medians, rather than the median of pairwise ratios. Its exact seeded percentiles are reproduced. The follow-up decision is data-dependent, so predeclaring the decision rule does not by itself establish 95% coverage for the final percentile intervals. The initial 15-pair sample is displayed for all groups, not only for early-stopped winners. Final intervals are descriptive and are not adjusted for selection, serial dependence or multiple comparisons.

An additional sensitivity diagnostic resamples within the initial and follow-up phases, keeping their respective sizes 15 and 16. Its interval position relative to one agrees with the archived pooled diagnostic in all eight extended groups. This check addresses the fixed phase composition, not adaptive stopping or all dependence. For $P_5/M6$ in H_5 at $d = 800$ the final intervals contain one; an unpassed two-IQR filter is not generally equivalent to such inclusion. No numerical significance level is claimed for the two-IQR rule.

10.4 Exact resources and numerical validation

Table S24: Total resources including the terminal residual. “Active counts outer cycles using an inherited predictor; “solves/MV separates triangular solves from additional matrix–vector products. Reference work uses the canonical weights, not calibrated multiprecision instruction times.

Field	d	method	cycles	active	F	J/LU	solves/MV	scales	work
H1	50	P_5	2	1	11	2/2	11/0	0	25000
H1	50	$M6$	2	0	7	4/2	10/4	80	22960
H1	200	P_5	3	2	16	3/3	17/0	0	37080
H1	200	$M6$	3	0	10	6/3	15/6	120	33820
H1	800	P_5	4	3	21	4/4	23/0	0	49160
H1	800	$M6$	4	0	13	8/4	20/8	160	44680
H1	1024	P_5	4	3	21	4/4	23/0	0	49160
H1	1024	$M6$	4	0	13	8/4	20/8	160	44680
H5	50	P_5	2	1	11	2/2	11/0	0	7328600
H5	50	$M6$	2	0	7	4/2	10/4	800	7111800
H5	200	P_5	3	2	16	3/3	17/0	0	10949800
H5	200	$M6$	3	0	10	6/3	15/6	1200	10604600
H5	800	P_5	4	3	21	4/4	23/0	0	14571000
H5	800	$M6$	4	0	13	8/4	20/8	1600	14097400
H5	1024	P_5	4	3	21	4/4	23/0	0	14571000
H5	1024	$M6$	4	0	13	8/4	20/8	1600	14097400

Table S25: Total resources including the terminal residual. “Active counts outer cycles using an inherited predictor; “solves/MV separates triangular solves from additional matrix–vector products. Reference work uses the canonical weights, not calibrated multiprecision instruction times.

Field	d	method	cycles	active	F	J/LU	solves/MV	scales	work
H1	50	P_7	2	1	15	2/2	15/0	0	31560
H1	50	$M8$	2	0	7	4/4	8/0	960	26760
H1	200	P_7	3	2	22	3/3	23/0	0	46920
H1	200	$M8$	3	0	10	6/6	12/0	1440	39520
H1	800	P_7	3	2	22	3/3	23/0	0	46920
H1	800	$M8$	3	0	10	6/6	12/0	1440	39520
H1	1024	P_7	3	2	22	3/3	23/0	0	46920
H1	1024	$M8$	4	0	13	8/8	16/0	1920	52280
H5	50	P_7	2	1	15	2/2	15/0	0	7993400
H5	50	$M8$	2	0	7	4/4	8/0	81600	12285800
H5	200	P_7	3	2	22	3/3	23/0	0	11947000
H5	200	$M8$	3	0	10	6/6	12/0	122400	18365600
H5	800	P_7	3	2	22	3/3	23/0	0	11947000
H5	800	$M8$	3	0	10	6/6	12/0	122400	18365600
H5	1024	P_7	3	2	22	3/3	23/0	0	11947000
H5	1024	$M8$	3	0	10	6/6	12/0	122400	18365600

Table S26: Final 1300-digit iterate reevaluated independently at 2600 digits. The floor flag concerns relative-precision censoring, not failure of the residual stop. Extension trajectories equal the corresponding initial trajectories exactly at both precisions; their repeated checks are preserved in the data.

Field	d	method	cycles	1300/2600	checked residual	checked error	floor
H1	50	P_5	2/2		1.0118×10^{-72}	2.5335×10^{-73}	no
H1	50	$M6$	2/2		4.1728×10^{-64}	9.9401×10^{-65}	no
H1	200	P_5	3/3		4.7011×10^{-493}	1.2045×10^{-493}	no
H1	200	$M6$	3/3		2.9827×10^{-390}	7.2474×10^{-391}	no
H1	800	P_5	4/4		6.7330×10^{-1301}	1.6970×10^{-1301}	yes
H1	800	$M6$	4/4		1.0073×10^{-1300}	2.4563×10^{-1301}	yes
H1	1024	P_5	4/4		6.7330×10^{-1301}	1.6970×10^{-1301}	yes
H1	1024	$M6$	4/4		1.0073×10^{-1300}	2.4563×10^{-1301}	yes
H5	50	P_5	2/2		3.4160×10^{-105}	8.6223×10^{-106}	no
H5	50	$M6$	2/2		4.3225×10^{-93}	1.0778×10^{-93}	no
H5	200	P_5	3/3		3.0148×10^{-716}	7.5864×10^{-717}	no
H5	200	$M6$	3/3		6.2622×10^{-568}	1.5880×10^{-568}	no
H5	800	P_5	4/4		2.5362×10^{-1299}	6.3713×10^{-1300}	yes
H5	800	$M6$	4/4		2.5372×10^{-1299}	6.3738×10^{-1300}	yes
H5	1024	P_5	4/4		2.5362×10^{-1299}	6.3713×10^{-1300}	yes
H5	1024	$M6$	4/4		2.5372×10^{-1299}	6.3738×10^{-1300}	yes

Table S27: Final 1300-digit iterate reevaluated independently at 2600 digits. The floor flag concerns relative-precision censoring, not failure of the residual stop. Extension trajectories equal the corresponding initial trajectories exactly at both precisions; their repeated checks are preserved in the data.

Field	d	method	cycles	1300/2600	checked residual	checked error	floor
H1	50	P_7	2/2		2.3569×10^{-127}	5.8076×10^{-128}	no
H1	50	$M8$	2/2		5.3349×10^{-115}	1.4162×10^{-115}	no
H1	200	P_7	3/3		2.2181×10^{-1125}	5.5584×10^{-1126}	no
H1	200	$M8$	3/3		1.1761×10^{-927}	2.9336×10^{-928}	no
H1	800	P_7	3/3		2.2181×10^{-1125}	5.5584×10^{-1126}	no
H1	800	$M8$	3/3		1.1761×10^{-927}	2.9336×10^{-928}	no
H1	1024	P_7	3/3		2.2181×10^{-1125}	5.5584×10^{-1126}	no
H1	1024	$M8$	4/4		6.7330×10^{-1301}	1.6970×10^{-1301}	yes
H5	50	P_7	2/2		6.8855×10^{-185}	1.7064×10^{-185}	no
H5	50	$M8$	2/2		9.6097×10^{-157}	2.4548×10^{-157}	no
H5	200	P_7	3/3		2.5362×10^{-1299}	6.3713×10^{-1300}	yes
H5	200	$M8$	3/3		3.1848×10^{-1266}	8.0460×10^{-1267}	yes
H5	800	P_7	3/3		2.5362×10^{-1299}	6.3713×10^{-1300}	yes
H5	800	$M8$	3/3		3.1848×10^{-1266}	8.0460×10^{-1267}	yes
H5	1024	P_7	3/3		2.5362×10^{-1299}	6.3713×10^{-1300}	yes
H5	1024	$M8$	3/3		3.1848×10^{-1266}	8.0460×10^{-1267}	yes

All 48 stored gates passed the recorded rule. Their cycles, resources, error norms and available COC quotients were independently recomputed from the saved dyadic iterates. The final 1300-digit iterates were independently reevaluated at 2600 digits using row-wise sums of the stated field formulas; every requested tolerance is met. This is a numerical check, not an interval-arithmetic certificate. The guard-floor comparison is the same as in the family campaign. A floor-censored final is not evidence of relative error accuracy and computed zero finals are excluded from COC.

For each extended group, initial and AMD follow-up histories are identical at both precisions. The 16 nominal paired configurations give 11 distinct exact paired paths: $P_5/M6$ repeats $d = 800/1024$ in each field; $P_7/M8$ repeats $d = 200/800$ in H_1 and $d = 200/800/1024$ in H_5 . Repeated paths are not additional nonlinear problems. Full errors, residuals, COC flags and dyadic vectors are retained; no asymptotic-order estimate is inferred from a short or precision-floor-terminated path.

10.5 Selected Intel follow-ups retained separately

Table S28: Earlier selected Intel follow-ups: 16 pairs per group. Absolute medians are seconds. The last two columns show the separate AMD ratios for comparison. These Intel observations are not pooled into any AMD aggregate, and the selected subset is not an independent test across all problems.

Pair	field	d	Intel A	Intel B	Intel ratio	AMD initial	AMD follow-up
$P_5/M6$	H1	50	0.270199	0.271059	0.996827	1.026028	1.028329
$P_5/M6$	H1	200	0.717279	0.694589	1.032666	1.033197	1.031889
$P_5/M6$	H1	800	0.457838	0.467143	0.980081	0.964340	0.971790
$P_5/M6$	H1	1024	0.667637	0.817844	0.816337	0.970887	0.971035
$P_7/M8$	H1	50	0.320742	0.300683	1.066709	1.061722	1.058703
$P_5/M6$	H5	200	141.262379	158.020287	0.893951	1.029712	1.028583
$P_5/M6$	H5	800	182.900488	182.680507	1.001204	1.000215	1.004053
$P_5/M6$	H5	1024	180.215876	179.778895	1.002431	1.004144	1.005537

These 256 earlier Intel timings concern the eight groups selected from the initial phase. They are not pooled with the AMD observations and do not replace any current result. Their saved trajectories agree with the corresponding initial ones, but timing ratios need not agree across platforms: notably, $P_5/M6$ in H_5 at $d = 200$ has an Intel follow-up ratio near 0.894 while the AMD phases are above one. Thus these follow-ups are a documented platform sensitivity, not evidence of uniform robustness or a controlled causal estimate of hardware effects.

10.6 Files, identity and remaining scope

The current received files are preserved under `data/multiprecision/external/current`. The reader-facing implementation is in `experiments/multiprecision/external`; summaries, phase statistics, allocation checks, and platform records are in `data/multiprecision/external/audit`. Separate Intel follow-ups are kept under `data/multiprecision/external/platform_sensitivity/intel_followups` and are excluded from the final aggregates.

The numerical audit has 7654 checks with no failure, including 48 precision gates and all 736 individual timing records. It does not re-run timed solvers or certify the mathematical results. The archived protocol identifier is

Protocol: `df85fadcce460b646664f6eba03773e05ad3c2bb5bda48d6f6ccc84a340b6075`.

Native Windows path separators are reconstructed for code-hash verification; source bytes are unchanged.

The external campaign is complete for these fields, distances and tolerances. It supplies neither the outstanding multiprecision fused- q timing battery nor a measured decomposition of interpreter, arithmetic and memory overhead. All multiprecision data described here are included in the repository; none is represented as part of the earlier Zenodo v1.0.0 software record.

11 Additional model diagnostics and detailed summaries

The following tables and figures complement the article and use its notation. Equation references point to the main article. The summaries retain all configurations, including unfavorable and unresolved timing comparisons.

11.1 Threshold sensitivity

Table S29: Sensitivity of the stationary threshold of $\hat{P}_N$ versus $P_N \circ P_N$. The first favorable integer dimension is shown for several conventions of κ .

N	1	1.2	1.298	1.7	3.23511	4.45444
2	55	54	54	54	52	51
3	97	96	96	96	94	93
4	151	150	150	150	148	147
5	217	216	216	216	214	213
6	295	294	294	294	292	291

11.2 Recorded environment, elementary weights and structural checks

Table S30: Environment preserved in the Hammerstein and $\hat{P}_2/P_{10}$ records. The 16 threads are the reported size of each pool, not the number of threads actually active in each call.

Item	Recorded value
Processor	AMD64 Family 25 Model 80 Stepping 0, AuthenticAMD
System	Windows-11-10.0.26200-SP0
Software	Python 3.13.6; NumPy 2.3.2; SciPy 1.17.1
Linear algebra	OpenBLAS 0.3.30, <code>pthread</code> layer; two libraries, associated with NumPy and SciPy, with pools of 16 threads
Cycle limit	MAXIT=20, recorded for both batteries

Table S31: Elementary weights expressed in equivalent products. Division retains the parameter κ ; the other weights are those used to account for the scalar functions in the fields.

Arithmetic	$x \times y$	x/y	$x^{1/2}$	$\exp x$	$\ln x$	$\sin x$	$\cos x$	$\arctan x$
Model	1	κ	3.0	16	14	14	14	14

Table S32: Structural audit of $H_1, \dots, H_5$ over the three distances $\delta = 0.10, 0.30, 0.60$. The rank and density columns show the observed minimum; for κ_2 and r_3 , the interval from minimum to maximum is shown.

Field	n	min rank J	min rank O	min rank($J - A_n$)	min. density	$\kappa_2(J) / r_3(O)$
H_1	20	20	20	20	1.000	1.404–1.411 / 0.753–0.754
H_2	50	50	50	50	1.000	1.483–1.484 / 0.892–0.893
H_3	100	100	100	100	1.000	1.488–1.488 / 0.944–0.944
H_4	150	150	150	150	1.000	1.495–1.497 / 0.963–0.963
H_5	200	200	200	200	1.000	1.493–1.494 / 0.971–0.971

11.3 Flatness and practical minimizers

Table S33: Quantification of the flatness of the optimum of P_N for $\kappa = 1$. The curvature χ_P is dimensionless.

Field	N_P^*	x_P^*	$\mathcal{B}_P(0.5\%)$	$\mathcal{B}_P(2\%)$	χ_P
H_1	3	2.50775	$\{3\}$	$\{2, 3\}$	7.51×10^{-2}
H_5	9	9.19358	$\{8, 9, 10, 11\}$	$\{7, \dots, 13\}$	3.67×10^{-3}

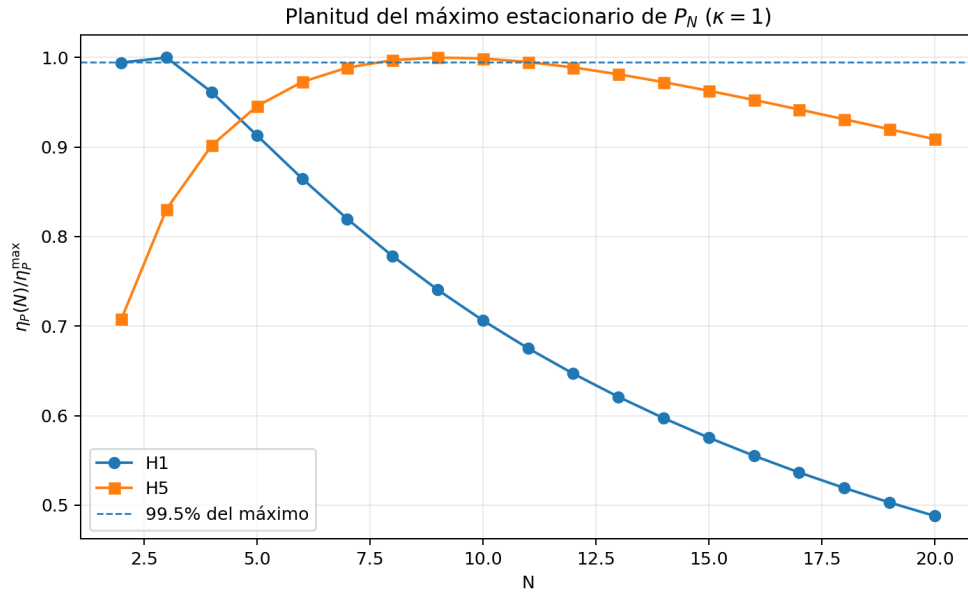


Figure S3: Normalized stationary index of P_N for $\kappa = 1$. The dashed line represents 99.5% of the maximum. In H_1 , only P_3 belongs to that band; in H_5 , the band contains P_8, P_9, P_{10}, P_{11} .

Table S34: Practical minimizers obtained by the complete sweep. The “1 cycle” column gives the first index of each family that meets the tolerance in the initial cycle; in that cycle P_j and S_j execute exactly the same block. Times are medians with IQR in parentheses, in milliseconds. Work corresponds to $\kappa = 1$ and is used only as an algebraic reference.

Field	δ	1 cycle	best P_N	wall time P	best S_m	wall time S	work P/S
H_1	0.10	5	P_5 (1)	0.1188 (0.0047)	S_5 (1)	0.1201 (0.0041)	12920/12920
H_1	0.30	7	P_2 (2)	0.1429 (0.0040)	S_2 (2)	0.1360 (0.0084)	15160/14760
H_1	0.60	9	P_2 (2)	0.1381 (0.0119)	S_2 (2)	0.1293 (0.0040)	15160/14760
H_5	0.10	4	P_4 (1)	1.0738 (0.1373)	S_4 (1)	1.0549 (0.1281)	3541200/3541200
H_5	0.30	5	P_5 (1)	0.9886 (0.0410)	S_5 (1)	0.9992 (0.0678)	3707400/3707400
H_5	0.60	6	P_6 (1)	1.0562 (0.1191)	S_6 (1)	1.0743 (0.1335)	3873600/3873600

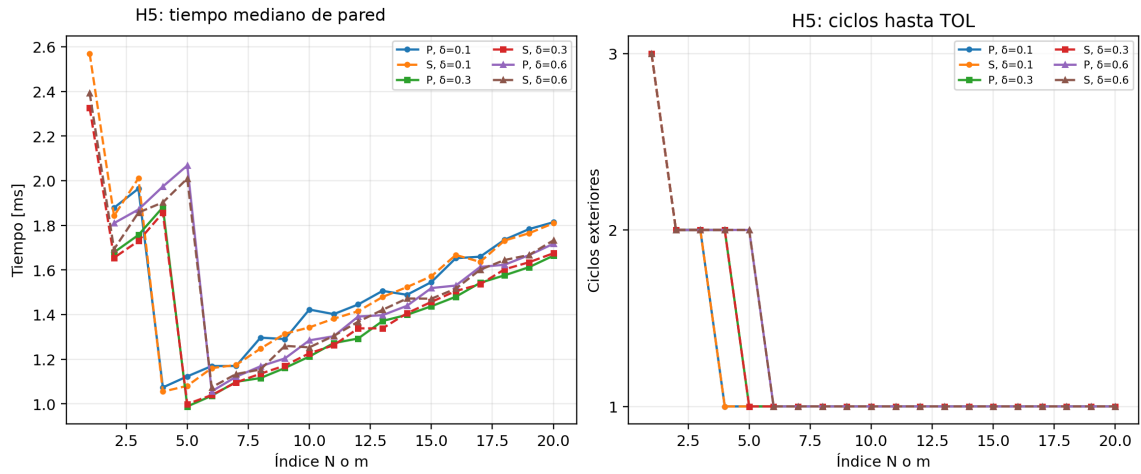


Figure S4: Complete sweep in H_5 for three initial distances. Left: median elapsed wall time. Right: number of outer cycles required to reach $TOL = 10^{-12}$. The first cycle of P_N and S_N has the same resources. With more than one cycle, memory is active, but that activation does not imply a resolved timing separation; the curves are interpreted together with the IQRs in this supplement.

11.4 Fusion resource counts and complete-solve summaries

Table S35: Exact resource counts for the fused comparison. Values in parentheses correspond to $N = 2$. The “end-to-end” row includes the common terminal evaluation of F used by the stopping criterion.

Regime	Method	F	J	LU	Solves	Matrix–vector products
Stationary	$\hat{P}_N$	$2N$ (4)	2	1	$(N+1)(N+3)$ (15)	$(N+1)^2$ (9)
	$P_N \circ P_N$	$2N$ (4)	2	2	$2(N+1)$ (6)	0
First D macrocycle, before the residual	$\hat{P}_N$	$2N$ (4)	2	1	$N^2 + 3N + 1$ (11)	$N(N+1)$ (6)
	$P_N \circ P_N$	$2N$ (4)	2	2	$2N + 1$ (5)	0
End-to-end, one macrocycle, with terminal residual	$\hat{P}_N$	$2N + 1$ (5)	2	1	$N^2 + 3N + 1$ (11)	$N(N+1)$ (6)
	$P_N \circ P_N$	$2N + 1$ (5)	2	2	$2N + 1$ (5)	0

Table S36: Practical end-to-end comparison of $\hat{P}_2$ versus $P_2 \circ P_2$. Work includes the common terminal evaluation of F listed in table S35. Negative values favor fusion.

n	Median ΔW [%]	Median Δt [%]	Range Δt [%]	clear differences/18
40	−2.5	+30.9	[+23.2, +35.2]	4/18
55	−10.1	+24.1	[+18.3, +26.5]	7/18
100	−23.3	+12.2	[+8.7, +15.0]	0/18
200	−34.6	−29.4	[−30.9, −27.6]	16/18

Table S37: Secondary nearby-order comparison of $\hat{P}_2$ and P_{10} . Define $\Delta W = 100(W_{\hat{P}_2}/W_{P_{10}} - 1)$ and $\Delta t = 100(t_{\hat{P}_2}/t_{P_{10}} - 1)$; negative values favor $\hat{P}_2$. The timing columns summarize the nine combinations of δ and tolerance per field.

Field	n	$10^6 \eta_{\hat{P}_2}/10^6 \eta_{P_{10}}$	ΔW [%]	median Δt [%]	range Δt [%]	clear/9
H_3	100	3.32012/3.17369	−12.35	+1.31	[−2.20, +3.62]	0/9
H_5	200	0.571324/0.554943	−8.67	−0.76	[−4.07, +1.41]	0/9

11.5 Consecutive truncation parameters

Table S38: Measured COC for $q = N + 1$ and $q = N + 2$ in the coupled polynomial stress field used for order verification.

N	q	reference order	observed COC
2	3	10.89897949	10.87757519
2	4	11.65685425	11.67053004
3	4	19.16515139	19.15817198
3	5	20.80776406	20.81728526
4	5	29.31782106	29.31487833
4	6	31.87450787	31.88067604